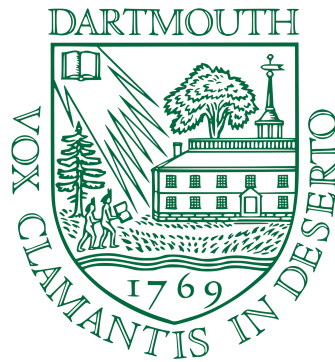


Simulation-Guided Prioritization and Implementation of Social Loafing Interventions in CS Group Projects



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Abstract

Social science research that compares candidate interventions has a structural problem: every comparison requires its own controlled study, and controlled studies are slow, expensive, and limited in scope. This thesis proposes that generative agent-based models (GABMs) can serve as a *prioritization layer* in the workflow; it will serve as a step inserted between literature review and human-subject experimentation that uses simulation to narrow a large set of candidate interventions down to the one or few most worth testing with real participants.

The thesis contributes a six-step framework for using GABMs as a prioritization layer and a proof of concept applying the framework to social loafing in computer science group projects, from literature review to an implementable tool.

The proof of concept proceeds as follows. A systematic review of the social loafing literature identifies six candidate interventions with empirical support: Agile methodologies, immediate and meaningful feedback, peer evaluation, task visibility, tracking individual contributions, and weekly logs. Using Google DeepMind’s Concordia library, 100 computer science teams of five agents are simulated, going through seven conditions: the control plus the six interventions, for a total of 700 simulations. Agents in the simulations are defined by eight dimensions: the Big Five personality traits, preference for group work, winning orientation, and skill level. An OLS regression with cluster-robust standard errors and Benjamini-Hochberg correction reveals that all six interventions significantly reduce perceived social loafing relative to the control condition, with immediate and meaningful feedback emerging as the most effective. This finding is translated into **Frank**, a Chrome extension that I built that overlays the GitHub interface to promote immediate, specific, and actionable pull request feedback. The next steps on **Frank** are to test it in the real world for validity.

Together, these contributions demonstrate that generative agent-based simulations are a viable and practical way to prioritize social interventions before investing time and resources in large-scale human-subject studies.

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Chapter 1

Introduction

1.1 The Problem

Social science research has a structural pacing problem. When a phenomenon has many plausible interventions, the field’s standard tool for comparing them, the controlled human-subject study, takes months to design, recruit for, and analyze; participant pools are often narrow, such as psychology undergraduates, single-gender samples, or single institutions [1]; and the technical and financial costs of running such studies at scale are substantial [2]. The result is that researchers have to choose, often on the basis of intuition, which of many candidate interventions to invest in testing, leaving promising candidates to be unevaluated for years. The field accumulates fragmented evidence about isolated interventions rather than systematic comparisons across them.

This problem is not unique to any one phenomenon, but it is especially relevant in domains where the settings change faster than research cycles. Technology-supported teams are a clear example: communication mediums, collaboration platforms, and work norms evolve continuously, and the antecedents of group dysfunction in these settings, such as anonymity, reduced social presence, the dehumanizing effects of text-based communication [3, 4], are difficult to replicate in laboratory conditions. By the time that a traditional research program has compared a handful of interventions, the setting it studied may no longer exist.

1.2 The Opportunity

Generative agent-based models (GABMs) offer a way to address this pacing problem without replacing human-subject research. A GABM populates a simulated environment with agents whose behavior is generated by a large language model, allowing researchers to model complex social interactions at a fraction of the cost and time of recruiting real participants [5, 6]. Recent work has shown that GABMs can produce behaviorally plausible group dynamics, replicate collective patterns observed in real social systems [7], and respond meaningfully to interventions injected into the simulation environment [2].

1.3 Roadmap

The thesis is organized in three parts.

Part I (Chapters 2–5) develops the framework. Chapter 2 provides background on agent-based models, generative agent-based models, and the cost problem in social science research. Chapter 3 addresses the foundational question of whether LLM-based simulations are representative enough of real human behavior to support the framework’s claims. Chapter 4 presents the six-step framework itself. Chapter 5 discusses the framework’s benefits and limitations and the conditions under which it is and isn’t appropriate to apply.

Part II (Chapters 6–10) is the proof of concept. Chapter 6 provides background on social loafing and explains why computer science group projects are a consequential setting in which to study it. Chapters 7–9 walk through the framework’s six steps as applied to this case: literature review and candidate selection, generative simulation, and comparative evaluation and prioritization. Chapter 10 introduces Frank, the tool that operationalizes the prioritized intervention for real-world testing.

Part III (Chapters 11–13) reflects on the case study. Chapter 11 discusses limitations specific to this application, distinct from the framework-level limitations covered in Chapter 5. Chapter 12 outlines next steps, including testing Frank in a real classroom and applying the framework to other domains. Chapter 13 concludes.

Part I

The Framework

Chapter 2

Background

2.1 The Cost Problem in Social Science Research

As discussed in the introduction, there is a structural pacing problem that has well-documented mechanics in the methodological literature. This section unpacks some of the specific cost problems in social science research as they pertain to the framework presented in Chapter 4.

Recruitment and Statistical Power

Comparing interventions requires enough participants to detect effects at the desired level of statistical power. For typical social science effect sizes, this often translates to dozens or hundreds of participants per condition. Recruitment is consistently identified as one of the most expensive parts of the research pipeline: it requires institutional review board (IRB) approval, participant compensation, scheduling infrastructure, and ongoing attention to dropout and replacement [2]. A study comparing two interventions may need 200 participants; one comparing more may need a considerable amount more. The cost scales roughly linearly with the number of conditions, while the value of each additional comparison may not.

Population Constraints

The participants who are easiest to recruit are not necessarily the participants most representative of the population the research aims to describe. Convenience sampling is widespread in social science research and is a known threat to generalizability [1]. A finding established with one population may not transfer to others, and the cost of replicating studies across populations compounds the problem. Researchers facing this trade-off often choose between a single study with high internal validity and limited generalizability, or a research program of comparable size that spreads across multiple populations and takes much longer to complete.

Time

Even when financial resources are available, time is rarely compressible. A controlled study can take six months to a year from approval to results when delays in recruitment, data collection, and review are accounted for. A sequence of studies comparing several interventions can therefore span half a decade or more before a comparative picture begins to emerge. For phenomena whose underlying conditions change on shorter timescales than

that, the comparative picture may be out of date by the time it is complete.

2.2 Agent-Based Models

Agent-based modeling (ABM) is a long-established computational technique for studying complex systems through simulation. In an agent-based model, individual actors (called agents) are defined by a set of attributes and behavioral rules, and the model observes how their interactions produce system-level outcomes over time. ABMs have been used productively across economics, epidemiology, urban planning, and ecology, and they have a particular appeal for social science because they make it possible to study emergent phenomena (segregation patterns, market dynamics, disease spread) that arise from the interactions of many individuals rather than from any single individual’s behavior [5].

The key limitation of traditional ABMs is that agent behavior is programmed by the researcher, based on assumptions or simplified rules [2]. This restricts the model’s ability to capture the full range of behaviors that arise in real-world situations. An agent whose response to a given situation is encoded as an if-then rule will only ever respond in the ways the researcher anticipated; any behavior outside that set, by construction, will never happen. For social science questions that depend on flexible reasoning, contextual interpretation, or social nuance, this rule-based limitation can produce simulations that capture the structure of an interaction without capturing what makes it psychologically real.

2.3 Generative Agent-Based Models

Generative agent-based models (GABMs) are an extension of ABMs, where the researcher’s rules for the agents are replaced by large language models that are given a personality and situation [2, 5]. Rather than encoding behavior as a fixed set of rules, the researcher describes each agent’s traits, goals, and current situation, and the language model generates their actions and responses at each step of the simulation. This shifts the source of behavioral variation from the researcher’s anticipated rule set to the language model’s reasoning, which allows agents to produce behaviors the researcher did not explicitly program.

GABMs have already been applied across a range of domains. In economics, they have been used to simulate market behavior and consumer decisions; in social science, they have been used to study group dynamics, opinion formation, and emergent social structure; in human-computer interaction research, they have been used to generate synthetic user responses for early-stage experimental design [5, 8]. The methodology is still young, and its validity is the subject of active debate (taken up in Chapter 3), but its capacity to model behaviorally-varied agent populations at low cost is already well established.

Several libraries and frameworks have emerged to support GABM research. Google

DeepMind’s Concordia, the library used in this thesis, provides infrastructure for designing GABM experiments with agents whose actions are grounded in physical, social, or digital contexts [6]. Concordia supports the design, execution, and analysis of complex simulations involving many interacting agents, and is structured to make experimental control and reproducibility tractable at scale.

2.4 Why GABMs Fit Social Science

Three properties make GABMs particularly well-suited as a tool for social science research.

Scalability. A GABM can simulate hundreds or thousands of agents at a small fraction of the cost of recruiting an equivalent number of human participants. For research questions that depend on observing many parallel groups under comparable condition, this changes what is feasible and accessible. A study that would have required years of recruitment can be run in days or weeks.

Behavioral diversity. Because each agent in a GABM is described in natural language, the researcher can vary agents along as many dimensions as the research question requires. Personality traits, goals, motivations, skills, prior experiences, demographic factors, and domain-specific dispositions can all be expressed and randomized independently. This heterogeneity is methodologically essential: simulations populated by homogeneous agents tend to produce artificially clean results that do not reflect the variation found in real groups [9]. GABMs make it possible to introduce the kind of behavioral diversity that real-world group dynamics require, in a way that traditional rule-based ABMs cannot.

Computational efficiency. The arrival of low-cost, low-latency language models (models optimized for high throughput rather than maximum capability) has made it financially viable to run simulations involving thousands of agent reasoning steps. The Gemini 2.5 Flash model used in this thesis is one such model, designed specifically for the kind of high-volume, real-time inference that large simulations require [10]. Without this generation of inference-optimized models, GABM research would be confined to far smaller experiments.

Scalability, behavioral diversity, and computational efficiency are necessary conditions for the kind of research this thesis proposes, but they are not sufficient. A simulation can be large, varied, and cheap, and still fail to produce findings that are meaningful for understanding real human behavior. The question of whether GABMs are representative enough of real-world dynamics to support the framework’s claims is the subject of Chapter 3.

2.5 The Big Five Personality Traits

A recurring requirement in agent design is the need for a principled way to introduce heterogeneity into the agent population. The Big Five personality taxonomy is one of the most widely used tools for this purpose, both in human psychological research and, increasingly, in generative agent work.

The Big Five identifies five broad dimensions along which individuals consistently differ [11]. Each dimension is treated as a continuum rather than a category, and an individual’s profile is described by their position along all five. A central strength of the framework is its capacity to capture personality at a broad level of abstraction without forcing individuals into discrete types, which makes it a well-established integrative model for personality research [11]. Below is a brief description of each of the five personality traits:

Openness to experience reflects how willing someone is to seek out new experiences and engage in learning [12]. It manifests in traits such as curiosity, imagination, and broadmindedness [12].

Conscientiousness reflects how dependable, hard-working, achievement-oriented, and persevering someone is [12].

Extraversion reflects how sociable, assertive, and communicative someone is [12].

Agreeableness reflects how cooperative, trusting, flexible, and tolerant someone is [12].

Neuroticism reflects emotional instability, including tendencies toward anxiety, worry, and insecurity [12].

The Big Five has been validated extensively. It has been replicated across cultures, age groups, and languages, and is generally regarded as one of the most empirically supported taxonomies in personality psychology [11]. This validation matters for simulation design: when a researcher introduces agent heterogeneity along Big Five dimensions, they are drawing on a taxonomy with substantial evidence that the dimensions correspond to real, recurring patterns of individual difference rather than to artifacts of any particular measurement instrument.

The Big Five has also been used in prior generative agent research as a basis for agent differentiation. Modeling agents with varied trait profiles produces populations whose collective behavior reflects realistic variation, and which respond differently to identical situations in ways that resemble the variation observed among real human participants. The framework presented in Chapter 4 prescribes the Big Five as a baseline starting point for agent design across social science domains, supplemented by phenomenon-specific dimensions drawn from the literature on the question being studied.

Chapter 3

Are Simulations Representative of the Real World?

Before presenting the framework, it is worth addressing a foundational question: can LLM-based social simulations produce findings that are meaningful for understanding real human behavior? The deeper question behind this one is older than LLMs. Joshua Epstein’s 1999 articulation of *generative social science* argued that agent-based computational modeling constitutes a third mode of explanation alongside induction and deduction, which is to explain a macroscopic social regularity means to show that it can be grown from the bottom up, by situating heterogeneous agents in an environment, letting them interact under local rules, and observing what emerges [13]. Epstein’s motto is: *if you didn’t grow it, you didn’t explain it*. It locates simulation not as a replacement for empirical human-subject work but as a distinct way of probing whether a proposed mechanism is even capable of producing the phenomenon of interest. LLM-based simulation inherits this role and extends it: where classical agent-based models required hand-coded behavioral rules, LLM agents bring more human-like reasoning to the same generative project [14].

The use of LLM agents in social simulation has grown rapidly, with applications in economics, education, game theory, and social networks [9]. This growth has produced a polarizing debate. Optimists argue that LLM-based simulations offer a scalable, low-cost alternative to traditional human-subject research [9]. Skeptics question whether LLMs can serve as valid proxies for human behavior at all, noting that ad hoc simulation techniques can introduce systemic biases that distort findings, particularly in high-stakes domains such as election forecasting [9]. A third, conditional position holds that LLM-based simulations are feasible but require careful structural and methodological constraints: behavioral mimicry alone is insufficient, and the cognitive structures and interaction mechanisms of agents must be thoughtfully designed [9]. This chapter takes the conditional position.

3.1 The Alignment Problem

The most relevant challenge for this thesis is what Wu et al. call the alignment problem, which is whether simulated agents’ behaviors and collective dynamics align with real societal phenomena [9]. Alignment in this sense used as a narrower empirical question of whether what happens inside a simulation tracks what would happen outside it. A simulation that

scores well on one notion of alignment can fail on the other: an agent population can be perfectly inoffensive and still be a poor model of how real students actually behave in a group project, and conversely an agent population can reproduce realistic but unflattering group dynamic, precisely because it has not been sanitized.

Wu et al. further argue that the alignment problem cannot be settled with a single measurement [9]. It must be examined at two levels: individual and collective. At the individual level, the question is whether a single agent behaves in a human-like way: does the agent’s stated preferences, reasoning, and reactions resemble those of the kind of person it is meant to represent? At the collective level, the question is whether agent interactions produce realistic social dynamics and emergent phenomena [9]. Both levels matter, but it is important to note this distinction.

A further distinction, sharpened by Lu et al. (2025), cuts across both levels, which is the difference between believability and accuracy [15]. A behavior is believable if a human observer cannot easily tell it apart from a real human behavior; it is accurate if it matches what a real human would actually do in the same situation. The early generative agents literature [16] was largely evaluated on believability, such as whether the simulated town of agents produced plausible-looking daily routines, conversations, and social events. Believability is necessary but not sufficient: an agent population can be entirely believable in its moment-to-moment behavior while still producing aggregate distributions that diverge sharply from real data.

3.2 Individual vs. Collective Alignment

Mou et al.’s survey of LLM-based social simulation makes the distinction between individual and collective alignment structurally precise. They categorize existing simulations into three types: (1) individual simulation, which mimics specific individuals or demographic groups; (2) scenario simulation, where multiple agents collaborate to achieve goals within a specific context; and (3) society simulation, which models interactions within agent societies to reflect the complexity of real-world dynamics [17]. These types form a progression rather than a partition. A scenario simulation is built out of individual simulations; a society simulation is built out of scenario simulations. Validity at a lower level is necessary but not sufficient for validity at a higher one.

Collective alignment in turn depends critically on agent heterogeneity, which is whether the population of agents exhibits sufficient behavioral diversity to generate the kind of complex, emergent dynamics seen in real social systems [9]. When agents respond heterogeneously to similar situations, their interactions produce feedback loops and emergent patterns that more closely mirror real group behavior [9]. A homogeneous population, on the other hand, overfits; its members converge on the same response to the same stimulus, and the resulting

simulations produce artificially clean results that do not reflect the messiness of real human groups. The emphasis on heterogeneity also resonates with Epstein’s original conception of generative social science, in which emergence at the macroscopic level is explicitly attributed to the local interaction of heterogeneous agents rather than the aggregation of representative ones [13]. Heterogeneity is the structural constraint that makes the conditional position of [9] achievable in practice.

3.3 Evidence from Adjacent Work

The empirical case that LLM agents can serve as proxies for individual human behavior rests on several converging lines of evidence at different behavioral grain sizes. The most ambitious is Park et al.’s 2024 simulation of 1,052 real people [16]. Their agents, conditioned on qualitative interview data from each participant, replicated participants’ responses on the General Social Survey at 85% of the accuracy with which the same participants replicated their own answers two weeks later, and performed comparably in predicting personality traits and outcomes in experimental replications. The 85% figure is striking not because it is high in absolute terms, but because the ceiling for the task (a human predicting their own answers two weeks later) is itself far below 100%. The result establishes that LLM-based individual simulation is in the same accuracy regime as human test-retest reliability for at least one well-studied class of behaviors.

A complementary result, with a narrower scope but a sharper behavioral test, comes from Xie et al. (2024), who examined whether LLM agents exhibit human-like trust behavior in Trust Games drawn from behavioral economics [18]. They find that LLM agents do exhibit trust behavior in these games and that GPT-4 agents show high behavioral alignment with humans on the relevant measures [18]. Park et al. establish alignment on broad survey-style attitudes; Xie et al. establish it on a specific interactive behavior. Both are necessary preconditions for the kind of scenario simulation this thesis relies on.

Evidence at the collective level is more recent and more tentative, but points in a similar direction. Cheng et al. (2024), drawing on Schelling’s Nobel Prize-winning model of micromotives and macrobehavior, found that segregated social patterns emerge from LLM agent interactions even under simple behavioral settings [7]. Crucially, they found that regardless of the bias score an LLM received on standard benchmarks, the macrobehavior of the population remained similar [7]. This result matters here because it suggests that individual-level LLM biases (the model being too agreeable, too verbose, or too prone to consensus) do not necessarily invalidate collective-level findings. The macro-pattern can be robust to the micro-noise. Mou et al.’s survey reinforces this from the other direction [17]: across the scenario and society simulations they review, the methodologies that produce credible collective dynamics are those that pair LLM agents with structured interaction

mechanisms and heterogeneous initialization rather than relying on raw LLM behavior alone.

The evidence cuts the other way as well, and the most pointed counterweight comes from Lu et al. (2025), who distinguish believability from accuracy in LLM behavior simulation and document specific failure modes [15]. In particular, they note that prompt-only LLM simulations routinely distort opinion distributions and overestimate effect sizes in experimental manipulations by factors of two to ten, and they incorrectly predict the direction of effects between ten and thirty-two percent of the time [15]. These are not minor calibration issues. They imply that any quantitative claim drawn from an LLM-only simulation should be treated as essentially uncalibrated. What survives this critique is ordinal comparison: the question of whether condition A produces a larger effect than condition B, rather than the question of how large that effect is.

None of this means the simulations are a substitute for real-world experimentation. Hämmäläinen et al. (2023) found that GPT-3 can generate synthetic responses to HCI research questions that are believable and difficult to distinguish from real responses, but they conclude that synthetic data is best used to ideate and pilot new experiments, not to draw final conclusions [8]. Any finding derived from synthetic data, they argue, must ultimately be validated with real participants [8]. The finding is consistent across the literature: simulations can sharpen a research question, narrow a design space, or surface an unanticipated dynamic, but it cannot close the loop on a causal claim about humans.

3.4 Implications for the Framework

These threads converge on a position that is more useful for this thesis than either uncritical optimism or blanket skepticism. LLM-based simulations are most defensible when they are used to compare conditions rather than to estimate effect sizes, when their agent populations are heterogeneous enough to generate emergent dynamics, and when their results are treated as inputs to subsequent human-subject work rather than as substitutes for it.

Chapter 4

A Framework for Using GABMs as a Prioritization Layer

4.1 The Argument

Social science research that compares candidate interventions has a structural problem: every comparison requires its own controlled study, and controlled studies are slow, expensive, and limited in scope. The result is that promising interventions go untested for years, and the ones that do get tested are often chosen on the basis of researcher intuition rather than systematic comparison.

This chapter proposes that generative agent-based modeling can act as a prioritization layer in the social scientist’s workflow as a step inserted between literature review and human-subject experimentation that uses simulation to narrow a large set of candidate interventions down to the one or two most worth testing with real participants. The framework I propose does not replace human-subject research; it changes which human-subject studies get run.

The framework’s claim is not that the highest-ranked intervention will be the most effective in a real classroom. It is that the highest-ranked intervention is the one most worth testing first.

The framework draws on an established methodological lineage. The view that social systems can be productively understood by simulating the behavior of individual agents long predates LLMs, and existing methodological guides for generative agent-based experimentation provide much of the scaffolding this framework builds on [2]. What this framework contributes is not a new mode of simulation but a specific role for simulation within the social science workflow. This framework inherits the validation and reliability concerns of generative agent-based modeling generally [9], and is designed to take those concerns seriously rather than around them.

4.2 The Six-Step Framework

The framework consists of six steps, illustrated in Figure 4.1. Steps 1 and 2 are familiar from any literature-driven research project. Steps 3, 4, and 5 are the prioritization layer that this framework contributes. Step 6 is the human-subject study that the framework’s

output is designed to inform.



Figure 4.1. The six-step framework.

4.2.1 Step 1: Literature Review

The goal of the literature review is to inventory candidate interventions: concrete, named strategies that the literature claims will mitigate the phenomenon of interest. At this stage, the researcher should prioritize breadth over depth. The researcher should aim to identify as many distinct, plausibly-effective interventions as possible, even if the empirical evidence for any one of them is uneven.

Two questions guide the review. The first is what interventions have empirical support, even if limited; the framework’s role is to figure out which of these under-examined candidates is worth a larger investment. The second is what interventions have been proposed but never tested; many papers often propose interventions that no one has actually evaluated, and these are candidates too.

The output of Step 1 is a list of candidate interventions, each with a citation and brief description of how it might work in practice and a clear understanding of how it can be evaluated.

4.2.2 Step 2: Candidate Selection

The researcher filters candidates using three criteria:

1. **Empirical basis.** The intervention should have at least some grounding in prior work.

2. **Simulatability.** The intervention must be expressible as a change in the agents’ environment, goals, or constraints.
3. **Implementability.** If the goal is to translate findings into a real-world tool, the intervention must be feasible to deploy.

The first criterion, empirical basis, is a quality filter. Pure speculation belongs in human-subject research, not in a prioritization filter that is itself drawing on the existing literature; if the literature contains no evidence about an intervention, the framework has nothing to weigh it against.

The second criterion, simulatability, reflects the limits of simulation. Interventions that depend on physical embodiment, real-time biometrics, or other features that cannot be represented in a language-based simulation should be excluded, not because they are unimportant, but because the framework cannot evaluate them. Excluding them honestly is better than evaluating them poorly.

The third criterion, implementability, is necessary only when the framework’s output is intended to inform a real-world deployment, as it is in this thesis. An intervention that requires institutional changes the researcher cannot make is not a useful prioritization target. A researcher whose end goal is purely scientific (publishing a comparative study) can relax this criterion; a researcher whose end goal is a deployable tool should not.

The output of Step 2 is a shortened list of the list from Step 1 that meet these criteria. These are the interventions that will be tested.

4.2.3 Step 3: Generative Simulation

The third step is the core methodological contribution of the framework. Each shortlisted intervention is simulated under conditions held as constant as possible, allowing the resulting differences in agent behavior to be attributed to the intervention itself rather than to differences in setup. Three design choices structure this step: agent design, environment design, and intervention injection.

Agent Design

The framework’s external validity depends on agent heterogeneity. A simulation populated by identical agents will produce artificially clean results that do not reflect the messiness of real human groups. Recent work on the validity of LLM-based simulation argues that collective alignment (whether agent populations produce realistic emergent dynamics) depends critically on whether agents respond heterogeneously to similar situations [7, 9]. Without heterogeneity, agents overfit to the prompt and the resulting dynamics tell the researcher more about the prompt than about the phenomenon being studied.

The framework’s prescription for operationalizing heterogeneity is to identify the individual-

difference variables that the literature on the phenomenon treats as consequential, and to randomize agents along those dimensions. This is a literature-driven decision, not an intuitive one. Different phenomena will demand different agent dimensions: a study of loafing may require personality traits and motivational dispositions, while a study of online harassment may require political orientation and platform tenure.

A useful starting point for agent design is the Big Five personality taxonomy: openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism. The Big Five is one of the most extensively validated personality frameworks in psychology, with cross-cultural replication and broad acceptance as an integrative model for research on individual differences [11]. It has also been used in prior generative agent work, where personality has been used to differentiate agents and produce behaviorally varied populations. For most social science problems, the Big Five is appropriate as a baseline set of agent dimensions, supplemented by domain-specific dimensions identified through the literature review.

The Big Five is not a complete description of any individual, and the framework does not treat it as one. Trait rankings shift with age and gender, the dimensions correlate with cultural orientation, and the taxonomy does not capture domain-specific dispositions such as math anxiety, language confidence, or risk tolerance. A research-backed agent design will typically combine validated trait dimensions with phenomenon-specific dimensions drawn from the same literature review that produced the candidate interventions.

Environment Design

The simulation environment should reflect the real-world setting in which the phenomenon occurs, but only to the level of detail that affects the outcome of interest. The researcher does not need to simulate every aspect of a workplace, classroom, or online platform; they need to simulate the aspects that the literature identifies as relevant. Adding detail beyond that point increases computational cost without improving validity, and risks introducing artifacts that the simulation interprets but the phenomenon does not.

In practice, environment design involves three concrete choices: the setting (in-person, virtual, hybrid, or determined by agents themselves); the structural parameters that the literature treats as consequential (group size, communication medium, task interdependence); and the duration of the simulation, which should be long enough to allow the phenomenon to emerge but short enough to remain computationally tractable.

Intervention Injection

Each intervention must be expressed in a way that agents can act on. In Concordia-style simulations, this means injecting the intervention at multiple levels: the global premise that frames the simulation, the goals each agent is given, and the situational context each agent

reasons from on every step [6]. Multi-level injection is important because LLM agents do not reliably remember information that appears only once at the start of a long simulation. Persistent reinforcement is what makes the intervention behaviorally active rather than ornamental.

The framework’s prescription for intervention design is also that prompts be developed with the same care as the wording of a survey instrument. Small wording changes can produce substantial behavioral changes, and the researcher should pilot each intervention prompt before running the full simulation. A sensitivity analysis, or rewording the intervention several ways and confirming that the qualitative results are stable, is a reasonable safeguard, even when full replication is not feasible.

4.2.4 Step 4: Comparative Evaluation

Step 4 is where the framework departs most sharply from existing generative agent-based work. Most published generative simulations report descriptive or qualitative findings about whether agents exhibit a particular behavior. The framework instead treats the simulation as the data-generating process for a formal statistical comparison.

This requires three commitments. The first is a control condition: every intervention must be compared not only to other interventions but to a baseline simulation in which no intervention is active. Without a control, the researcher cannot distinguish intervention effects from baseline behavior. The second is an outcome measure drawn from the literature on the phenomenon, rather than invented for the simulation. Validated measures carry meaning that ad hoc measures do not, and they make the simulation’s findings comparable to existing human-subject evidence. The third is a statistical analysis appropriate for the nested data structure that simulations naturally produce. Agents are nested within groups, and groups are nested within conditions. Standard errors must reflect this nesting (typically through clustering at the group level), and multiple comparisons across interventions must be corrected (Benjamini-Hochberg is a defensible default).

The output of Step 4 is a ranked list of interventions, with effect sizes, confidence intervals, and control-intervention contrasts indicating which interventions outperform which others. This list is descriptive: it tells the researcher what the simulation found. It does not yet tell them what to do next.

4.2.5 Step 5: Intervention Prioritization

Step 5 is interpretive rather than computational. The researcher should use the Step 4 output and identify which intervention or interventions are most promising to test with human participants.

The researcher should weigh four considerations. The first is the size and statistical significance of the effect. The second is the interpretability of the effect: an intervention

that produces a smaller but well-understood effect may be more promising than one with a larger but harder-to-explain effect, because the former is easier to operationalize and easier to learn from in a follow-up study. The third is the cost and feasibility of testing the intervention with human subjects; an intervention that requires a year-long course redesign is more expensive to test than one that can be deployed as a single tool. The fourth, when the framework’s output is intended for deployment, is implementability as a real-world tool; an intervention whose simulation effect is strong but whose real-world implementation is impractical is not a useful prioritization target.

The output of Step 5 is one to a few prioritized interventions, with a concrete rationale explaining why each was selected and why the others were not. It should be clear how to implement this intervention in the real world.

4.2.6 Step 6: Human-Subject Study

Step 6 is not part of the simulation framework, but it is what the framework is designed for. The prioritized interventions from Step 5 are the input to a traditional randomized controlled trial with human participants. The framework’s contribution is that the researcher arrives at Step 6 with a smaller, better-justified set of interventions to test, rather than guessing which of fifteen candidates is most worth a year of recruitment.

4.3 What the Framework Is and Isn’t

It is a filter, not an oracle

The framework does not predict real-world effect sizes. It produces a rank ordering of interventions, and that ordering is only as good as the agent design, the prompts, and the underlying language model. The framework’s claim is not that the highest-ranked intervention will be the most effective in a real classroom; it is that the highest-ranked intervention is the one most worth testing first. This is the same logic that motivates pilot studies in clinical research, computational screens in chemistry, and agent-based economic models; none of which are taken to be substitutes for the empirical work they precede.

This distinction matters because it sets the bar the framework must clear. A predictive simulation must reproduce real-world effect sizes; a filtering simulation only needs to reproduce relative ordering well enough to be useful as a prioritization signal. Existing evidence from generative agent-based work suggests that the latter is more tractable than the former [7, 8], and the framework is built around that weaker, more defensible claim.

It can be used to form hypotheses, not just to rank them

The framework as presented takes a set of candidate interventions from the literature and ranks them. But the same machinery can be used in a generative direction, which is to

surface hypotheses that the literature has not yet articulated. A researcher who runs a heterogeneous agent population through a scenario simulation and observes the resulting dynamics is, in effect, watching a low-cost pilot study. Patterns that emerge are candidate hypotheses for human-subject testing in their own right.

This generalizes the framework’s role from intervention prioritization to hypothesis prioritization. The six steps remain the same in structure: a literature review establishes the relevant individual differences and outcome measures, a heterogeneous simulation generates behavior under varied conditions, comparative evaluation surfaces what is different across conditions, and the most promising patterns are carried forward into human-subject work. The difference is that the inputs to Step 2 (candidate selection) need not be a list of named interventions; they can also be a set of conditions, agent compositions, or environmental parameters whose effects the researcher is curious about but for which the literature offers no prior claim. The simulation’s output then identifies which of these conditions produces effects worth investigating empirically.

The same epistemic caveats apply. The simulation cannot confirm a hypothesis; it can only surface one and supply an initial reason that it is worth testing. A hypothesis generated this way still requires human-subject validation, and the framework’s role remains that of a filter; in this case, it functions as a filter on the space of possible questions rather than on the space of possible answers.

It does not handle interaction effects well

Real interventions are often combined; the framework tests interventions in isolation, and provides no information about how they interact when deployed together. Extending it to factorial designs is possible but expands the simulation cost considerably: testing six interventions in isolation requires seven conditions (six interventions plus control), while testing all pairwise combinations would require many more. Researchers whose primary interest is interaction effects should design accordingly from the start, accepting the additional cost.

Its validity depends on the underlying model

Different LLMs produce different agent behavior. Different versions of the same model produce different behavior. Findings from one model may not replicate under another, and the framework therefore requires that the researcher disclose which model was used and, ideally, replicate the prioritization step on at least one other model to confirm that the ranking is stable. Where cross-model replication is infeasible, the limitation should be acknowledged and the framework’s output should be treated as conditional on the specific model used.

This dependence is not unique to this framework; all generative agent-based research

inherits it, but it is more consequential here because the framework's output is a prioritization decision that downstream human-subject testing will rely on.

Chapter 5

Benefits and Limitations of the Framework

This chapter outlines the benefits and limitations of the framework presented in Chapter 4.

5.1 Benefits

5.1.1 Reduces Experimental Costs

The dominant cost of comparing candidate interventions through traditional means is the cost of running multiple human-subject studies. This includes recruitment, compensation, instrumentation, and the time of the researchers running them. Even modest social science experiments often take months to design, recruit for, and analyze, and the costs of doing so at scale are substantial [2]. Generative agent-based simulations shift this cost structure dramatically. A simulation of hundreds of teams across many conditions can be run for the cost of LLM inference compute, which is significantly cheaper than recruiting and compensating an equivalent number of human participants. For a researcher comparing six or more candidate interventions, this is the difference between a multi-year research program and a multi-week one.

The cost reduction is not just financial. It is also a reduction in the friction of running comparisons. A researcher who can simulate six interventions in two weeks is more likely to run those comparisons in the first place than one for whom each comparison is a year-long undertaking. The framework therefore makes broad, comparative, prioritization-oriented research feasible and accessible that previously was not.

5.1.2 Narrows the Intervention Space

A literature review on most social science phenomena produces many candidate interventions. A researcher facing a large set of candidate interventions has historically had two options: pick one or two interventions based on intuition and test them rigorously, or attempt to test all of them and produce shallow evidence about each. Neither is satisfying.

The framework offers a third option. Run all candidates through simulation, observe which produce the strongest effects under comparable conditions, and bring only the most promising one or two forward into human-subject testing. The set of interventions that reaches the

expensive testing stage is smaller and better-justified than it would be otherwise. This is the framework’s most concrete contribution: it converts a long list of plausible interventions into a short list of empirically-grounded priorities, before any human participants or money is involved.

5.1.3 Accelerates Behavioral Research

The time horizon of traditional intervention comparison is on the order of years. A controlled study takes months to run; a sequence of studies comparing multiple interventions, with the inevitable delays of recruitment, analysis, and publication, can take years to produce a meaningful comparison [1]. By the time the comparison is complete, the setting being studied may have changed significantly, and the most relevant interventions may no longer be the ones that were originally on the list. The framework does not eliminate the need for the slower, human-subject phase, but it shifts the slow work to a smaller and better-targeted set of questions.

5.1.4 Prioritizes Human-Subject Studies

One of the framework’s most important benefits, aside from making research cheaper and faster, is that it improves the quality of the prioritization decision that precedes every human-subject study. Traditionally, the question "which intervention should we test next?" has been answered by some combination of researcher intuition, publication trends, institutional pressure, and convenience. The framework offers a way to answer that question with systematic evidence, even when that evidence is itself imperfect.

This matters because human-subject testing is the bottleneck. The framework does not aim to replace human-subject research; it aims to make sure the human-subject research that does get conducted is targeted at the interventions most likely to produce meaningful results.

5.2 Framework-Level Limitations

The framework has limitations that are intrinsic to its approach rather than to any particular application of it.

Model Dependence

The framework’s output depends on the language model underlying the simulation. Different models (and different versions of the same model) produce different agent behavior, and the relative ranking of interventions could differ under a different model [10]. The framework requires that the model used be disclosed, and ideally that the prioritization step be replicated on at least one alternative model to confirm ranking stability. Where this is

not feasible, the framework’s output should be treated as conditional on the specific model used, and the limitation should be acknowledged explicitly. This is a real cost because it means that the framework cannot produce model-independent claims about the world, only model-conditional claims that point toward what to investigate further.

Prompt Sensitivity

The framework’s prescriptions about multi-level intervention injection (4) reduce but do not eliminate prompt sensitivity. Wording choices in the premise, agent goals, and situational context can drive behavioral differences that are not part of the intervention being tested. Sensitivity analysis, involving rerunning the simulation with several wordings of the same intervention, is a partial mitigation, but it is rarely conducted exhaustively in practice. The framework therefore inherits a degree of researcher discretion in prompt design that cannot be fully removed.

The Alignment Ceiling

The framework cannot exceed the alignment of the underlying simulation. If LLM agents are systematically misaligned with real human behavior in some way, for example, by overstating consensus-seeking, or by underweighting in-group bias, the framework will inherit that misalignment, and its prioritization output will reflect it. Recent work on the alignment of LLM-based simulations has been cautiously optimistic about collective-level dynamics but has also documented persistent gaps in individual-level accuracy [8, 9]. The framework’s response to this is structural rather than evidential: it positions its output as a filter for human-subject testing rather than a substitute for it, so that alignment gaps are caught at Step 6 rather than allowed to drive deployment decisions. But the dependence remains, and a future improvement in LLM alignment would directly improve the framework’s reliability.

No Native Support for Interaction Effects

The framework as described tests interventions in isolation. Each shortlisted intervention is simulated separately, and the comparative evaluation produces a ranking under the assumption that each intervention is the only one active. In practice, interventions are often combined: instructors do not deploy peer evaluation, contribution tracking, and feedback norms in isolation, and the effects of combinations may differ substantially from the effects of components. Extending the framework to factorial designs is possible but expands the simulation cost considerably. A researcher whose primary interest is interaction effects rather than ranking should design accordingly from the start, accepting the additional cost or restricting attention to a smaller set of candidate interventions.

Limited Longitudinal Capacity

Simulations that span long time horizons are expensive and accumulate context bloat; LLM context windows are finite. Phenomena that develop over months or years such as reputational dynamics, habit formation, accumulated trust, may be difficult to capture in a simulation that remains computationally tractable. The framework can be applied to longitudinal phenomena, but the researcher should expect to truncate the simulated time window and acknowledge that some longitudinal dynamics are not represented. This is one of the strongest reasons not to treat simulation output as a substitute for longer-running human-subject studies.

5.3 When the Framework Is and Isn't Appropriate

The framework is most appropriate when three conditions hold. (1) The phenomenon under study has multiple candidate interventions, drawn from a reasonably mature literature, that the researcher is choosing among. (2) The relevant individual differences and outcome measures are already identified by that literature, so that agent design and evaluation can be grounded in existing work rather than invented for the simulation. (3) The cost of running parallel human-subject studies for each candidate is high enough that prioritization meaningfully changes what gets tested.

The framework is less appropriate when these conditions are weak. Phenomena that depend heavily on embodiment, physical environment, biometric response, or other features that language-based simulation cannot represent are poor candidates: the framework can only simulate what can be expressed in agent goals, contexts, and interactions. Phenomena where the relevant individual differences are not yet identified by the literature force the researcher to guess at agent dimensions, which weakens the framework's collective alignment. Phenomena where the outcome measure is itself contested (where the field has not converged on what to measure) will produce a framework output that is hard to interpret, because the measure being optimized may not be the measure that matters.

The framework is inappropriate as a substitute for human-subject studies in any final-decision context. It is a filter, not an answer. A researcher who uses the framework's output to deploy an intervention without subsequent human-subject testing is over-interpreting the framework, and the framework's claims do not support that use. The framework's value lies in directing human-subject resources where they will have the most return, not in replacing those resources.

Part II

Proof of Concept: Social Loafing in CS Group Projects

Chapter 6

Background on Social Loafing

6.1 Engagement Theory and Group Work

In 1998, Kearsley and Shneiderman proposed the engagement theory as a framework for technology-based teaching and learning [19]. Engaged learning means that all student activities involve active cognitive processes such as creating, problem solving, reasoning, decision making, and evaluation [19]. The theory is grounded in the idea of creating successful collaborative teams that work on ambitious projects meaningful to someone outside the classroom [19]. In this collaborative setting, students need guidance on working together, encompassing skills such as project management, scheduling, time management, leadership, and consensus building [19]. One of the research questions the framework raises is: what kind of groupware, or collaborative software tools, would best support engagement theory? [19].

Collaborative group work holds that students can learn valuable lessons about communication and problem solving that transfer readily to professional environments [20]. Working in groups can significantly increase learning perceptions and problem-solving skills, and help students achieve a deeper level of understanding than individual work alone [20]. Group work is not without its drawbacks, however. It requires greater time investment and dependence on others, which can lead to counterproductive issues such as non-contributing members, unequal workload distribution, scheduling conflicts, and interpersonal tension [20].

6.2 Defining Social Loafing

Social loafing is the phenomenon that people working in groups exert less effort than they would individually [1]. Numerous studies have confirmed its existence through experiments involving rope-pulling, clapping, shouting, and other maximizing, unitary, and additive tasks.

Social loafing is related to, but distinct from, free riding. Free riding occurs when an individual fails to contribute their proportionate share of work yet still benefits from the group's output [20]. Both phenomena involve reduced effort, whether due to motivation or circumstance [20], but they differ in degree: social loafers do contribute to the final product, whereas free riders do not contribute at all [12].

The phenomenon was first observed in 1913 by Maximilien Ringelmann, a French

agricultural engineer, who found that when people collectively pulled on a rope, their combined output was less than the sum of their individual efforts [1]. Rope pulling is maximizing, unitary, and additive, meaning Ringelmann expected effort to scale directly with group size. Instead, he found that dyads pulled at 93% of their combined individual capacity, triads at 85%, and groups of eight at only 49% [21]. Ringelmann never published his results, and the findings were not revisited until 1974, when Ingham, Levinger, Graves, and Peckham replicated the experiment and showed that individual effort declines curvilinearly as group size increases, even when participants merely believed they were working with others [22]. This confirmed that the effect was driven not only by coordination loss but by motivation loss, and that increases in group size are inversely related to individual performance [22]. Ingham et al. later named this the phenomenon “social loafing” [1].

Latané et al. regard social loafing as a social disease, noting that it has negative repercussions for individuals, social institutions, and societies [21]. It leads to a reduction in human efficiency, which results in lowered productivity and benefits for everyone involved [21]. The behavior is also self-reinforcing: those who engage in social loafing are more likely to repeat it in future group settings [20]. Despite these drawbacks, Latané et al. recognize the importance of group work for accomplishing what individuals cannot do alone [21], and suggest that the goal should be to channel social forces so that group membership intensifies individual responsibility rather than diffusing it [21].

6.3 Why CS Group Projects Are a Consequential Setting

Computer science group projects are both practically important and understudied as a setting for social loafing research. Software development is fundamentally team-based, and the patterns of behavior that students form in undergraduate group projects extend into the professional environments they go on to join. If social loafing is the phenomenon that group membership diffuses individual responsibility, then teaching students to recognize and resist it in college projects is also a form of professional preparation.

The setting is also distinctive in ways that intensify social loafing. Technology-supported teams introduce antecedents to loafing that are difficult to replicate in a laboratory setting [3]. Three properties matter most: reduced social presence, the uneven visibility of technical work, and the dehumanizing effects of text-based communication.

Communication technology reduces perceptions of social presence, which makes it harder to feel the presence of others and leads to a lack of awareness of others and their emotions [3]. The result is increased anonymity within the group, even among members who know each other personally. Anonymity is one of the most consistently documented antecedents of social loafing in technology-supported teams: when individual contributions cannot be readily attributed, the social sanctions that ordinarily constrain low-effort behavior weaken.

Members can “hide in the crowd” [21], exerting less effort with less risk of being noticed by either teammates or instructors. It has been proposed, without empirical evidence, that some of the antecedents to social loafing are magnified within virtual teams [23].

Computer science projects also produce artifacts that are unevenly visible to outside observers. The number of GitHub commits and pull requests a student submits is not necessarily proportional to the time and effort they have invested, and it does not reflect the quality of those contributions. A single well-considered architectural decision may take days of design work and appear in the commit history as a single small change; a student who spends an entire week debugging a tricky integration issue may have nothing visible to show for it. Time spent on design, debugging, testing, and code review is largely invisible to instructors and frequently invisible to teammates as well. This invisibility cuts both ways: students who contribute substantially through hard-to-see work can be perceived as loafing when they are not, and students who are loafing can sometimes obscure that fact through high commit counts on shallow work.

Finally, the dehumanizing nature of text-based communication has been empirically linked to increased loafing in technology-supported settings [3]. When communication strips out the cues that ordinarily signal another person’s humanity, such as facial expressions, tone of voice, body language, the moral weight of letting them down lightens. Monzani et al. (2014) found that the negative effects of perceived loafing on group outcomes were stronger in text-based groups than in face-to-face groups, and proposed that the natural mechanisms by which groups absorb and rationalize occasional low effort, including interpersonal attraction, member similarity, the ability to read a teammate’s face, are largely unavailable when collaboration is mediated by text [4].

These properties make computer science group projects a particularly consequential setting in which to apply the framework. The setting is one in which loafing is easy to sustain and difficult to detect, the candidate interventions are practically deployable through the platforms students already use, and the population affected (computer science students, who will go on to be software engineers) is large and professionally consequential. The framework is most valuable in settings where the problem is significant, the existing literature is sparse [23, 24], and the cost of running parallel human-subject studies is prohibitively high. Computer science group projects meet all three conditions.

Chapter 7

Applying the Framework: Literature Review and Candidate Selection

7.1 Systematic Review Methodology

The first step was a systematic review of the social loafing literature. This included 30 experiments spanning the years 1974 to 2025, of which six pertained specifically to technology-based group projects. Each study contributed evidence about the situational, interpersonal, or individual factors that influence social loafing. While going through the literature, proposal of interventions or the testing of interventions were noted as possible interventions for the simulations.

7.2 The Big Five Personality Traits and Social Loafing

As described in the framework outlined in 4, the simulations contain agents that have personalities and are aware of the situation. Part of the personalities that these agents embody is the Big Five. Detailed below are each of the Big Five's connections or impacts to social loafing based on existing literature.

Openness to experiences A group's average openness score has been found to be positively related to group performance, though the direct relationship between openness and social loafing is not significant [12]. There is reason to believe the relationship may become positive in monotonous task environments [12]. Sherif (2022) found that students consistently described social loafers in their groups as uncreative and closed to new ideas [25].

Conscientiousness It is a valid predictor of job performance across occupations [12], and Tan & Tan (2008) found it to be negatively related to social loafing [26]. However, in group settings where responsibilities are shared, individual accountability can be diluted, weakening this effect [27]. In the IT sector specifically, the relationship between conscientiousness and social loafing is insignificant, likely because conscientiousness predicts contextual performance more strongly than task performance, and social loafing in technical work is more closely tied to the latter [12]. Sherif (2022) found that peers most often described loafers as disorganized

and frequently unavailable [25].

Extraversion Extraverts are perceived by groupmates as having greater impact on group outcomes and tend to generate more ideas when brainstorming [12]. Extraversion has consistently been shown to be a significant predictor of social loafing [12]: when extraverts lack motivation, they may cause process loss rather than drive group performance [12]. Sherif (2022) found that faculty and peers alike described social loafers as reserved, quiet, and unenthusiastic [25].

Agreeableness Teams with more agreeable members tend to outperform those with fewer [12], though agreeableness is more strongly related to interpersonal facilitation than to task performance, making its relationship with social loafing negative but not significant [12]. Sherif (2022) found that loafers were often described as compliant but not altruistic, as social loafing is fundamentally a self-serving behavior [25]. A theme emerging from faculty interviews was that loafers tend to avoid conflict unless directly complained about, at which point blame-shifting becomes common [25].

Neuroticism Neurotic individuals may avoid collaborative responsibilities due to stress, impairing both interpersonal dynamics and task performance [27]. Neuroticism has been found to be a significant predictor of social loafing [12], as it shares characteristics with obstructionism, which is behavior that hinders one's ability to perform [12]. Interestingly, Sherif (2022) found that student-identified loafers tended to appear emotionally stable and laid-back rather than anxious, likely reflecting a lack of investment in the course rather than emotional dysregulation [25].

7.3 Factors That Encourage Social Loafing

7.3.1 Defensive Impression Management

Mulvey and Klein (1998) note that perceptions of reduced effort and actual reduced effort may covary, but social loafing can occur without other group members perceiving the reduced effort [28]. Impression management is defined as an attempt by an individual to influence how others perceive them [28]. Mulvey et al. cite Tedeschi and Reiss (1981) who note that individuals may engage in impression management for assertive or defensive reasons [28]. Defensive impression management strategies are typically used when poor individual performance occurs and can take the form of apologies, excuses, disclaimers, and others [28]. An example of this is a group member telling others that they are not very comfortable in front of an audience, so that they don't expect as much from them in an oral presentation [28]. Defensive impression management in a group is likely to influence perceptions of loafing within the group as they are more likely to make attributions about the lack of effort group members are exhibiting [28]. If group members are making excuses for their performance, frequently apologizing for not following through, or continuously warn others to not expect

as much from them, they might be perceived as not trying as hard as they can or loafing [28]. This may lead to other group members subsequently lowering their efforts and increasing their own use of defensive impression management behaviors [28].

In 1998, Mulvey et al. explored the effects of defensive impression management perceptions on two group effectiveness variables: satisfaction with team members and group performance [28]. Mulvey et al. look at the interactive effects between perceived coworker loafing and defensive impression management [28]. They hypothesized that if group members already perceive a high level of loafing within their group, the use of defensive impression management likely confirms and reinforces these perceptions [28]. In their experiment, they put participants in groups and ask them to consult an organization and write a report that reviewed, critiqued, and made recommendations about a managerial issue, such as conflict resolution [28]. They found that both defensive impression management and perceived loafing were negatively correlated with satisfaction with group members and group performance [28]. The moderating effect of perceived loafing on the relationship between defensive impression management and group performance was noteworthy as the interaction term accounted for an additional 19% of the variance in group performance [28].

7.3.2 Dehumanization

Dehumanization is defined as “the denial of qualities associated with meaning, interest, and compassion” toward others, which is salient in technology [3]. When someone does not perceive another person as having similar values, their perception of shared humanity decreases and they are more likely to disregard the interests of others [3]. In 2010, Alnuaimi et al. studied the effects of dehumanization in collocated versus dispersed teams of various sizes through brainstorming, and found that dehumanization was a significant antecedent of social loafing [3].

Additionally, in 2014, Monzani et al. researched the effects of low fidelity computer mediated communication, in the form of text messaging, on perceived loafing [4]. They conducted an experiment by placing organizational psychology students in project groups, who worked on a project in four sessions over the course of one month [4]. They found that the negative effect of perceived loafing is stronger in low fidelity computer mediated work [4]. Monzani et al. pose a possible explanation for this: face-to-face groups, member similarity, and interpersonal attraction can be used as substitutes to rational accounts for loafing behaviors [4]. The example given is a loafer using their natural charm to justify their free-riding behavior, which is harder to do when you cannot see someone’s face or emotions [4].

7.3.3 Dispersion

Dispersion is when members of a team are located in different physical locations and communicate electronically [3]. Studies comparing collocated and dispersed teams have shown that collocated teams outperform dispersed teams, measured as per person number of generated ideas [3]. Increased dispersion is associated with increased anonymity, decreased visibility, and difficulty of making social comparisons, which all contribute to an increase in social loafing [3].

7.3.4 Group Size

In 1979, Latané et al. discussed how the theory of social impact explains social loafing [21]. The theory of social impact is based on the premise that if someone is the target of social forces, increasing the number of people in a group should therefore decrease the amount of pressure on each individual because the impact is divided amongst more people [21]. A greater diffusion of responsibility limits the exercise of self-control by obscuring autonomy for the team outcome [3]. In situations where pressures to work are external and individual outputs are not identifiable, the division of impact should lead each individual to work less hard [21]. Therefore, whether an individual is dividing up the amount of work they think should be performed or whether they are dividing up the amount of reward expected to gain, they will work less hard in groups, and therefore leads to a decrease in output [21]. Latané et al. conclude that the theory of social impact suggests that the amount of effort spent on group tasks should decrease as an inverse power function of the number of people in the group [21].

Further, in 1984, Zaccaro studied the interaction between task attraction and group size [29]. Zaccaro split participants, who were University of Connecticut psychology students, into two groups: a high task attractiveness group, where participants were told about the importance of the study and received extra credit for their class, and a low task attractiveness group, where they weren't informed about the importance of study nor given extra credit for their class [29]. Zaccaro found that while performance decreased as group size increased in the low task attractiveness group, performance increased as group size increased in the high task attractiveness group [29]. As group size increases, it becomes more difficult for individuals to encourage and monitor one another; this leads to a lack of feedback and support that further contributes to social loafing [30].

Research suggests that there is no significant gain in group sizes larger in size, with the optimal small group size being five [20]. Piezon and Donaldson cite Hare (1981) who suggested that group satisfaction becomes an issue for even numbered groups because of the development of subgroups, making five the optimal group size [20].

7.3.5 Perceived Dispensibility of Effort

In 1983, Kerr and Bruun explored the effects of group size and member ability on free-rider behavior [31]. In their first experiment, they used a 3 (group sizes of 2, 4, and 8) X 2 (high, low member ability) X 3 (additive, conjunctive, and disjunctive task demands) X 2 (subject sex) factorial design, where participants were asked to blow as much air as they could through a mouthpiece in 30 seconds [31]. They found that for disjunctive tasks, members with low ability felt that their efforts were more dispensible, and therefore exerted less effort; similarly, for conjunctive tasks, members with high ability felt more dispensable and therefore exerted less effort [31]. While perceived dispensability of effort does lend itself to social loafing for disjunctive and conjunctive tasks, it does not result in any net process loss to the group [31].

7.3.6 Task Interdependence

In 2003, Liden et al. explored individual-level antecedents of social loafing, one factor being task interdependence [30]. Their definition of task interdependence is consistent with that of Shea and Guzzo's (1987): "the degree of task-driven interaction among work group members" [30]. Liden et al. conducted surveys and interviews with employees at participating organizations. They found that when the perception of task interdependence is high, individuals may believe that their work is indistinguishable from the effort put forth by their coworkers, and may reduce their effort, concluding that task interdependence is positively related to social loafing [30].

7.4 Factors That Deter Social Loafing

Factors marked with * were used in the experiment.

7.4.1 Agile Methodologies*

In 2006, McAvoy and Butler studied social loafing in Agile teams [32]. Agile methodologies involve encouraging groups to take ownership over their code [32]. The basic principles of Agile are:

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

McAvoy and Butler (2006) posit that one of the unintended consequences of Agile Software Development is that it induces social loafing, under certain conditions [32]. To test

this proposition, they conducted an experiment on two software development teams over the source of eight months to see if the values inherent in Agile Software Development lead to social loafing [32]. They found that the project team that fully adopted Agile methodologies were less likely to exhibit social loafing [32]. Agile software development methodologies promote higher levels of social cohesion and a sense of ownership among team members [32].

7.4.2 Distributive Justice

In 2003, Liden et al. explored individual-level antecedents of social loafing, one factor being distributive justice [30]. Distributive justice, in organizational behavior literature, is the belief that one is being compensated according to their worth, in an effort to motivate individuals to put in effort [30]. Liden et al. cites George (1995) who found that social loafing was lowest under conditions of contingent rewards, which is consistent with organizational justice theory [30]. Therefore, individuals may reduce their effort when they consider the group size, task visibility, and perceived loss of wages if dismissed [20]. Liden et al. conducted surveys and interviews with employees at participating organizations and confirmed past research that distributive justice is negatively related to social loafing [30].

7.4.3 Expectations of Partner Performance

In 1985, Jackson and Harkins hypothesized a possible explanation for social loafing, where people working in groups expect their co-performers to loaf, so they reduce their own efforts to establish an equitable division of labor [33]. The experiment consisted of participants screaming alone and in pairs, subsequently answering questions about how much effort they put in when alone versus in a pair, how much effort their partner put in when alone versus in a pair, and how good they and their partner were at the shouting task [33]. Their results showed that when participants were informed that their partner intended to work hard all of the time, there was no alone/group difference [33].

7.4.4 Fatigue

In 1998, Hoeksema-van Orden et al. tested the effects of fatigue on social loafing [34]. They conducted two experiments with 64 male students who worked almost continuously for 20 hours under various social conditions [34]. In the first experiment, participants were split into two groups: individual and group conditions [34]. In the second experiment, all participants worked in groups [34]. They concluded that fatigue induces social loafing [34]. They also found that by individualizing the task and giving public individual feedback seemed to counteract the effects of social loafing [34].

7.4.5 Group Cohesiveness

Group cohesiveness is defined as the degree to which members are attracted to one another and choose to stay together [30]. Many studies have been conducted to examine the association between cohesiveness and social loafing; social loafing has been found to only occur in groups that were not cohesive or mildly cohesive, with highly cohesive groups showing members working equally hard collectively as they did individually [30].

In 1984, Zaccaro examined the hypothesis that internal group pressures to maximize productivity would moderate social loafing [29]. Zaccaro found that the interaction of the group helps apply these internal pressures to produce and that identifiable individual effort focuses the application of the pressures to produce [1]. Group cohesiveness encompasses group interaction, task commitment, and identifiability, which are all possible deterrents of social loafing [1].

In 2014, Fang and Chang tested the impacts of team member familiarity on social loafing in technology-supported teams [35]. Using a brainstorming experiment with university students in Taiwan, they found that the task performance of the unfamiliar teams were significantly lower than those of the familiar teams [35]. This is due to familiar teammates being more understanding of one another, so they can predict their teammates' future behavior according to their previous experiences, which reduces a lot of distrust and uncertainty [35].

7.4.6 Immediate and Meaningful Feedback*

In 2005, Piezon and Donaldson presented an overview of the psycho-social aspects of social loafing in a traditional and distance learning environment [20]. Amongst this overview is a list of frequently cited antecedents and their mitigating factors for instructors, designers, and administrators to use in distance education [20]. One of their recommendations includes ensuring individuals receive immediate and meaningful feedback [20].

7.4.7 Measuring Individual Efforts*

Five years after the Ingham et al. experiment, in 1979, Harkins, Latané, and Williams replicated Ringelmann's experiment with cheering and clapping, later adding another variable: telling each participant that their individual efforts would be measured [1]. Their results showed that the belief that individual efforts are being measured does discourage social loafing.

In the same study, Harkins et al. aimed to understand if social loafing was a result of an allocational strategy, where people conserve their strength for individual trials, or a minimizing strategy, where people do the bare minimum to "get by" because they cannot be individually identified [36]. Their clapping and cheering study suggests that when participants cannot be rewarded or punished for the individual efforts, people will loaf [36].

7.4.8 Peer Evaluation*

In 2008, Aggarwal and O'Brien identified some easy-to-implement project set-up factors and examined their impact on reducing the incidence of social loafing [37]. These four factors were: scope of the project, group size, group selection, and use of multiple peer evaluations [37]. Peer evaluations can be a tool to make individuals accountable for their contributions to the group task [37]. They cite Clark (1989), who noted the validity and reliability of peer evaluations as measures of student contribution and discusses that a key benefit of peer evaluations is that it puts pressure on peers to contribute, reducing work inequity and conflict [37]. In their study, Aggarwal and O'Brien found a negative and statistically significant relationship between the number of peer evaluations and the incidence of social loafing [37].

7.4.9 Perceived Coworker Loafing

Perceived coworker loafing is the extent to which group members feel one or more of their coworkers are social loafing [30]. Liden et al. cite a study by Schnake (1991) that proposed the "sucker effect" where group members withhold their own effort when they believe that others are withholding effort [30]. Contrary to Liden et al.'s hypothesis, perceived coworker loafing is negatively correlated with social loafing [30]. Essentially, individuals attempt to compensate for the lowered output of coworkers by working even harder either for their own personal gain (success of the group benefits them) or for the altruistic reason that their efforts to compensate will benefit their fellow coworkers [30].

7.4.10 Preference for Group Work*

Preference for group work refers to the degree that individuals prefer group work and compare it favorably to autonomous work [38]. It can be thought of as an attitude with a structure comprising affective and cognitive components in classic attitudinal terms, where the evaluation of group work elicits certain behavioral tendencies and intentions that may lead to a consistent pattern of proscribed patterns [38]. A positive evaluation of group work might lead to higher social loading because these individuals believe that their efforts are dispensable [38]. A study from 1995 by Wagner found a strong positive relationship between preference for group work and shared responsibility [38]. In 2007, Stark et al. conducted an experiment with groups of students in a business administration course and found that preference for group work is negatively related to self-reported and peer-related measures of social loafing [38].

7.4.11 Self-Evaluation

In 1987, Harkins and Szymanski tested whether self-evaluation was enough to reduce social loafing [39]. Their results showed that self-evaluation to a social standard given to individuals performing a maximizing task, in this case brainstorming, was all the motivation needed in order to motivate an individual not to practice social loafing, individually and with a partner [39].

A year later, in 1988, Harkins and Szymanski tested if an individual's self-evaluation with an objective standard would have the same results as the social standard. Self-evaluation provides the participants with two incentives: they can learn something about how well they can perform this task and they have the opportunity to take pleasure from beating a goal (through surpassing others or by surpassing a benchmark) [39]. There is this psychological factor of self-validation, or less cynically, self-knowledge, where individuals don't just want to see how they compare to others, but they want to learn that they are better than others [39]. Harkins and Szymanski sought to determine whether the opportunity to compare one's performance against an objective standard was sufficient to motivate performance [39]. Their experiment involved participants pressing a button whenever they saw a dot flash on a TV screen, split into eight groups: (1) output feedback, standard, experimenter evaluation, (2) output feedback, standard, no experimenter evaluation, (3) output feedback, no standard, experimenter evaluation, (4) output feedback, no standard, no experimenter evaluation, (5) no output feedback, standard, experimenter evaluation, (6) no output feedback, standard, no experimenter evaluation, (7) no output feedback, no standard, experimenter evaluation, and (8) no output feedback, no standard, no experimenter evaluation [39]. They concluded that the opportunity to compare one's performance to an objective standard is sufficient to motivate performance [39].

7.4.12 Task Visibility*

In 2003, Liden et al. explored individual-level antecedents of social loafing, one factor being task visibility [30]. Task visibility is one's belief that a supervisor is aware of their efforts [30]. If they are perceived as "hiding in the crowd," individuals may see no benefit in exerting high effort and no punishment in exerting low effort [21]. Individuals working on a collective task will lead to a decrease in self-awareness, which results in a disregard of performance standards and engagement in less self-regulation [20]. Liden et al. conducted surveys and interviews with employees at participating organizations [30]. They found that task visibility is negatively related to social loafing [30].

7.4.13 Winning Orientation*

Winning orientation, or individual differences in self-aggrandizement, relates to the strength of an individual's desire to win, especially in terms of maintaining superiority or favorable comparisons to social contexts [38]. Stark et al. cite Franken and Prpich (1996) who argued that individuals who have a high winning orientation are driven by ego concerns and strongly endorse self-image protection issues as the foundation of their competitive behavior [38]. Social loafing behavior can exact a significant cost in terms of their image in the social context, making those with a high winning orientation less likely to social loaf [38]. In 2007, Stark et al. confirmed this hypothesis with groups of students in a business administration course, concluding that winning orientation is a significant predictor of peer-rated loafing [38].

7.4.14 Weekly Logs*

In 2005, Piezon and Donaldson reviewed antecedents and mitigating factors for social loafing in traditional and distance learning environments, providing a set of recommendations for instructors, designers, and administrators [20]. Among these recommendations was the use of a weekly progress log as a structural mechanism for maintaining individual accountability across the lifespan of a group project [20]. The proposed log includes three components: outlining project milestones, tracking group and individual progress against those milestones, and defining hard completion dates for outstanding tasks [20].

7.5 Selected Candidate Interventions

The interventions selected for simulation were chosen on two criteria: (1) they had at least some empirical basis in the literature, and (2) they were feasible to simulate and plausible to implement. This yielded six candidate interventions: Agile methodology, providing immediate and meaningful feedback, tracking individual contributions, peer evaluations, increasing task visibility, and weekly logs.

7.6 Hypotheses

Based on prior empirical evidence, it is predicted that all six interventions will produce statistically significant reductions in social loafing relative to the control condition.

H_{01} : Agile methodology has no significant effect on reducing social loafing in computer science group projects.

H_{A1} : Agile methodology significantly reduces social loafing in computer science group projects.

H_{02} : Immediate and meaningful feedback has no significant effect on reducing social loafing

in computer science group projects.

H_{A2} : Immediate and meaningful feedback significantly reduces social loafing in computer science group projects.

H_{03} : Peer evaluation has no significant effect on reducing social loafing in computer science group projects.

H_{A3} : Peer evaluation significantly reduces social loafing in computer science group projects.

H_{04} : Task visibility has no significant effect on reducing social loafing in computer science group projects.

H_{A4} : Task visibility significantly reduces social loafing in computer science group projects.

H_{05} : Tracking individual contributions has no significant effect on reducing social loafing in computer science group projects.

H_{A5} : Tracking individual contributions significantly reduces social loafing in computer science group projects.

H_{06} : Weekly logs have no significant effect on reducing social loafing in computer science group projects.

H_{A6} : Weekly logs significantly reduce social loafing in computer science group projects.

Chapter 8

Applying the Framework: Generative Simulation

The next step of the framework is to simulate the intervention candidates.

8.1 Conceptualization

8.1.1 Research Objectives

The goal of the simulations is to determine the most promising intervention(s) to mitigate social loafing in computer science group projects. From the social loafing literature, six different interventions were empirically-backed to have some leverage in mitigating social loafing: (1) Agile methodologies, (2) immediate and meaningful feedback, (3) peer evaluations, (4) increasing task visibility, (5) tracking individual contributions, and (6) weekly logs.

8.1.2 The Premise

The premise of the simulations is that a computer science group, defined by five agents, is assigned to build a working software system together for their final project for their class. They have four weeks to complete the project and the simulation begins with the group meeting after class to discuss the project.

8.1.3 Duration and Setting

This simulation takes place over the course of four weeks. The simulation begins at the start of week 7 of a ten-week-long computer science project.

8.2 Tool Selection: Concordia and Gemini 2.5 Flash

For my experiment, I used Google DeepMind’s Concordia library to simulate computer science group projects with Google’s Gemini 2.5 Flash. Gemini 2.5 Flash is optimized for high-speed, low-latency, and cost-effective real-time applications, making it a good fit for a simulation like this [10]. Gemini 2.5 Flash’s sparse Mixture-of-Experts (MoE) transformer architecture dynamically routes tokens to a small subset of specialized experts, which decouples parameter count from inference cost, allowing for the high knowledge capacity

of a large model, yet maintains the speed and efficiency of a smaller one [10]. MoE is the primary reason that Gemini 2.5 Flash has a low-cost, high-speed performance, and why it was chosen for this experiment [10]. To design the Concordia simulations, I use the framework outlined by García Navarro et al. (2024) for designing reliable experiments with generative agent-based modeling [2].

8.3 Agent Design

8.3.1 Modeling Dimensions

Each agent is intended to represent one member of a computer science group project. I define each member with the following eight characteristics: the Big Five personality traits: (1) openness, (2) conscientiousness, (3) extraversion, (4) agreeableness, and (5) neuroticism, (6) preference for group work, (7) winning orientation, and (8) skill level. The Big Five personality traits (openness, conscientiousness, extraversion, agreeableness, and neuroticism) are randomized to be one of three levels: low, medium, high for each of the five personality traits. The preference for group work and winning orientation are both randomized to be low, high. Lastly, the skill level is randomized to be low, medium, high.

8.3.2 The 2,916 Possible Agents

In total, there are $3^5 \cdot 2 \cdot 2 \cdot 3 = 2916$ possible combinations of these factors and therefore 2916 possible agents for simulations.

From these 2916 distinct agents, I randomly select five agents to be put on a computer science group project team, with replacement, to represent similar personalities in the real-world. Research suggests that there is no significant gain in group sizes larger in size, with the optimal small group size being five, therefore I limit group sizes in the simulations to strictly five members [20].

From all of these possible combinations, 100 teams of 5 agents are randomly-generated, with replacement, to be used for each of the control and interventions. While this is a small subset of the population, it is roughly the size of the populations used for the majority of studies conducted on social loafing.

8.3.3 Sub-Goals and Behavioral Tags

All agents are given the goal of helping their team complete the computer science group project successfully. Agents are also given sub-goals, or additional commands, based on their personalities. These are defined as follows:

Personality Attribute	Sub-Goal
-----------------------	----------

High winning orientation	You want your contribution to be visible and recognized
Low winning orientation	You care more about getting the project finished than competing with teammates
High preference for group work	You prefer active collaboration, discussion, and shared decision-making
Low preference for group work	You prefer independent work and minimal coordination overhead
High skill level	You are one of the technically strongest people on the team
Medium skill level	You are reasonably capable and can contribute steadily
Low skill level	You are less confident in your technical ability and may avoid difficult tasks

Agents are also given behavioral tags based on their personalities and combinations of these personalities. The behavioral tags are appended to each agent’s descriptions. These are encoded as follows:

Personality Attribute(s)	Behavioral Tag
Low conscientiousness and low preference for group work	Higher risk of disengaging from coordination
High conscientiousness and medium or high skill level	Reliable at completing assigned work
High extraversion and high winning orientation	Likely to dominate discussion
High agreeableness	Tries to preserve group harmony
High neuroticism	More reactive under time pressure
High openness and medium or high skill level	More willing to propose novel technical ideas

8.4 Intervention Protocols

Each intervention is injected at three levels: the premise, agent goal, and situational context. Full prompts are reproduced in Appendix 13. The first level is during the description of the premise, shown to the Game Master and all agents and seen from step 1. The second level is that the intervention is prepended to every agent’s goal so that every planning and action decision is framed by the condition. The third level is that the intervention is appended to the situational context that each agent reasons from on every step, making it a persistent feature of the agent’s reality.

8.4.1 Agile Methodologies

The last intervention is to apply Agile methodologies. Teams that fully adopt an Agile philosophy tend to have team members that work better as a group by supporting each other, rather than hiding within the group to disguise problems with their quality of work [32].

In the simulation, the team has fully adopted an Agile philosophy from week 1. Their work is organized in weekly sprints with a sprint backlog, a daily stand-up, a sprint review, and a retrospective. The Agile norm emphasizes collective ownership, mutual support, and transparency: team members are expected to surface blockers immediately in stand-ups rather than hiding problems, to help teammates who are stuck rather than working in isolation, and to hold each other accountable through retrospectives.

8.4.2 Immediate and Meaningful Feedback

One suggested intervention for minimizing social loafing is ensuring that individuals receive meaningful and immediate feedback [20].

In the simulation, the team agrees that every contribution receives meaningful timely feedback. Concretely, this means that code submitted to the shared repository is reviewed within 24 hours with specific, actionable comments (not just ‘LGTM’). This also means that design proposals are discussed in the same session they are raised, with a clear decision or next step recorded. Also, when a teammate is struggling or off-track, the team addresses it directly and constructively in the current session rather than letting it accumulate. Lastly, written feedback is given within 48 hours.

8.4.3 Peer Evaluation

Peer evaluations have been found to be valid and reliable measures of student contribution [37]. A key benefit of peer evaluations is that it puts pressure on peers to contribute, reducing work inequity and conflict [37]. Possible ways to implement a peer evaluation intervention is adopting early implementation, having multiple evaluation points, and having specific

evaluation criteria [20].

In the simulation, the team uses a structured peer evaluation system that gets implemented every week. Each peer evaluation uses specific evaluative criteria: contribution to tasks, communication quality, and reliability (meeting commitments). These scores are averaged across teammates and returned to each student with written comments. These peer evaluations contribute to individual project grades.

8.4.4 Task Visibility

Liden et al. conducted surveys and interviews with employees at participating organizations and found that task visibility is negatively related to social loafing [30]. Possible interventions of implementing task visibility are to establish performance targets, communication procedures, and methods for solving problems [20].

In the simulation, the team has established task-visibility norms. This includes: (1) performance targets: every task has a named owner, a clear quality definition, and a firm deadline visible to all, (2) communication procedures: team members post a brief update whenever they start, finish, or are blocked on a task, and (3) problem-solving methods: when a blocker arises, the owner flags it immediately and the team uses a defined escalation path (pair-help → group discussion → instructor) rather than staying stuck silently.

8.4.5 Track and Publicize Individual Contributions

The first intervention has to do with tracking and publicizing individual contributions. Studies have suggested that this will lead to a greater sense of personal achievement and therefore lead to reduced social loafing [20].

In the simulation, each team member's contributions are logged on a shared board visible to all teammates and course staff. The board records design decisions, code commits, test coverage, and documentation written by each person. An up-to-date entry is required from every member after each session. Visible contributions recognize effort and achievement; blank or spare entries are equally visible and are noted.

8.4.6 Weekly Logs

Maintaining a weekly log has been identified as a way to try to minimize social loafing [20]. This would include outlining milestones, tracking group and individual progress, and define hard completion dates [20].

In the simulation, the team is required to submit a structured weekly progress log every week. The log records: (1) project milestones and whether they were met, (2) each team member's individual progress and contributions that week, and (3) expected completion dates for all outstanding tasks. The agents are informed that these weekly logs are a part of their grade.

8.5 Measuring Social Loafing

Perceived social loafing was the chosen metric for measuring social loafing in the simulations. Perceived loafing is an appropriate outcome measure because it captures the group's lived experience of loafing regardless of whether actual effort reduction has occurred. As Mulvey and Klein note, perceptions of reduced effort and actual reduced effort may covary, but social loafing can occur without other group members perceiving it [28]. Conversely, if loafing is not perceived by other group members, reduced effort would not be expected to negatively influence group satisfaction or performance [28]. The perceptions of group members are therefore important in examining the consequences of loafing on the group, regardless of the accuracy of those perceptions [28]. In this way, perceived loafing is both what the measurement instrument captures directly and the dimension of loafing most consequential for group dynamics.

Perceived social loafing was measured using the Perceived Social Loafing Questionnaire (PSLQ) developed by Mulvey and Klein [40]. At the end of each simulation, each agent rated every other agent in their group on four statements using a five-point Likert scale. These statements are as follows:

1. Members of my group are trying as hard as they can. (R)
2. Members of my group are "free-loaders."
3. Members of my group are contributing less than I anticipated.
4. Given their abilities, my group members are doing the best they can. (R)

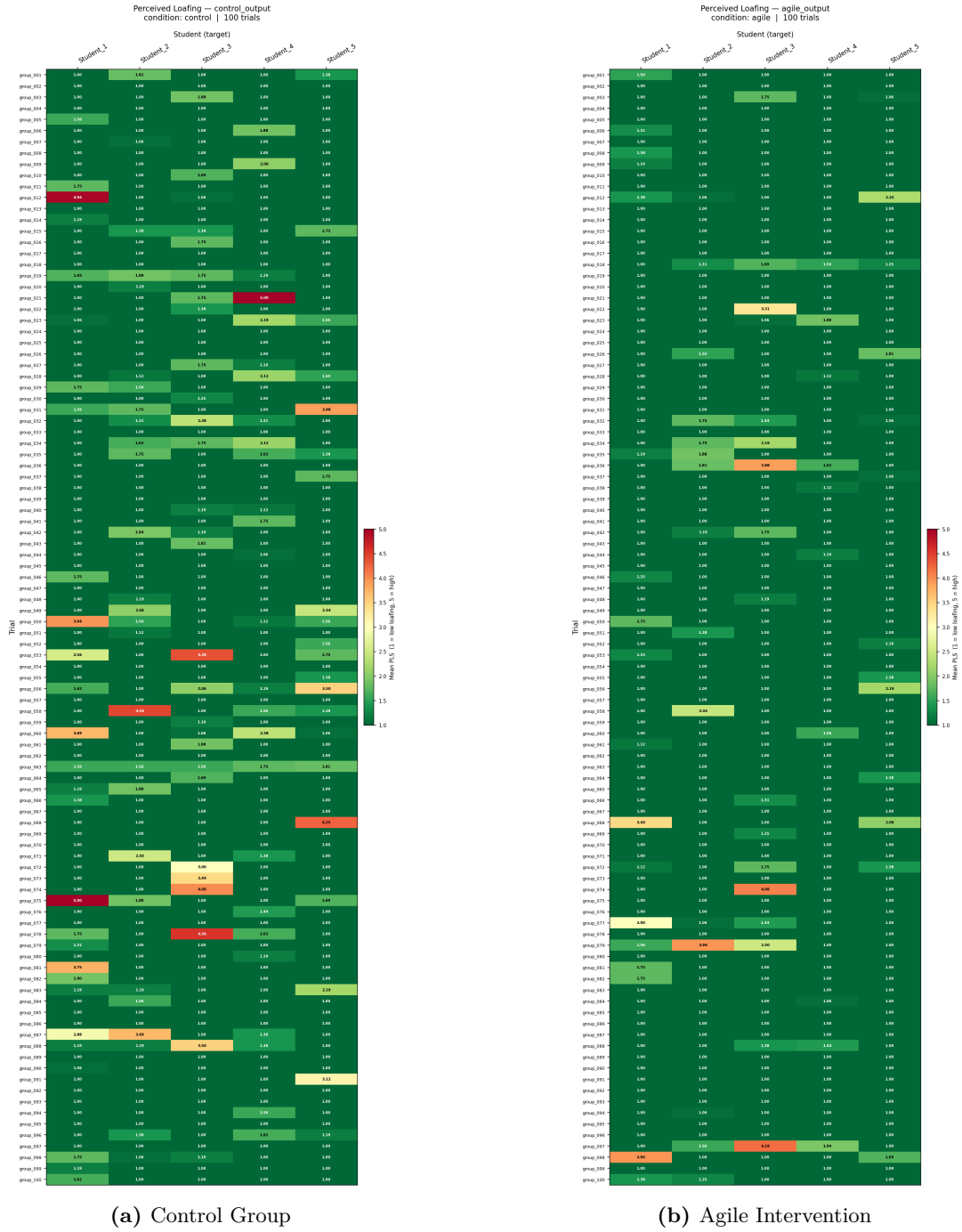
These ratings were averaged across raters to produce a perceived social loafing (PLS) score for each agent. A lower PLS score indicates less perceived loafing.

Chapter 9

Applying the Framework: Comparative Evaluation and Prioritization

9.1 Heatmap Output

Each heatmap below displays perceived social loafing (PLS) scores across 100 simulated trials for a given condition. Rows represent groups (group_001 through group_100) and columns represent the five students within each group (Student_1 through Student_5). Each group contains the same five agents across the control and six interventions. Each cell shows the mean PLS score assigned to that student by their four groupmates, where a score of 1 indicates no perceived loafing and a score of 5 indicates the maximum level of perceived loafing. The color scale runs from dark green (low perceived loafing, $PLS = 1$) through yellow-orange to red (high perceived loafing, $PLS = 5$). A predominantly dark green heatmap therefore indicates a condition in which group members perceived little loafing across most trials. Each intervention heatmap is shown next to the control heatmap to make direct visual comparisons.



Visually, the Agile intervention heatmap (Figure 9.1) appears notably greener than the control, with fewer and less intense orange cells, and zero red cells. This suggests that the Agile intervention reduces instances of extreme perceived loading, yet still produces some amount of perceived social loading.

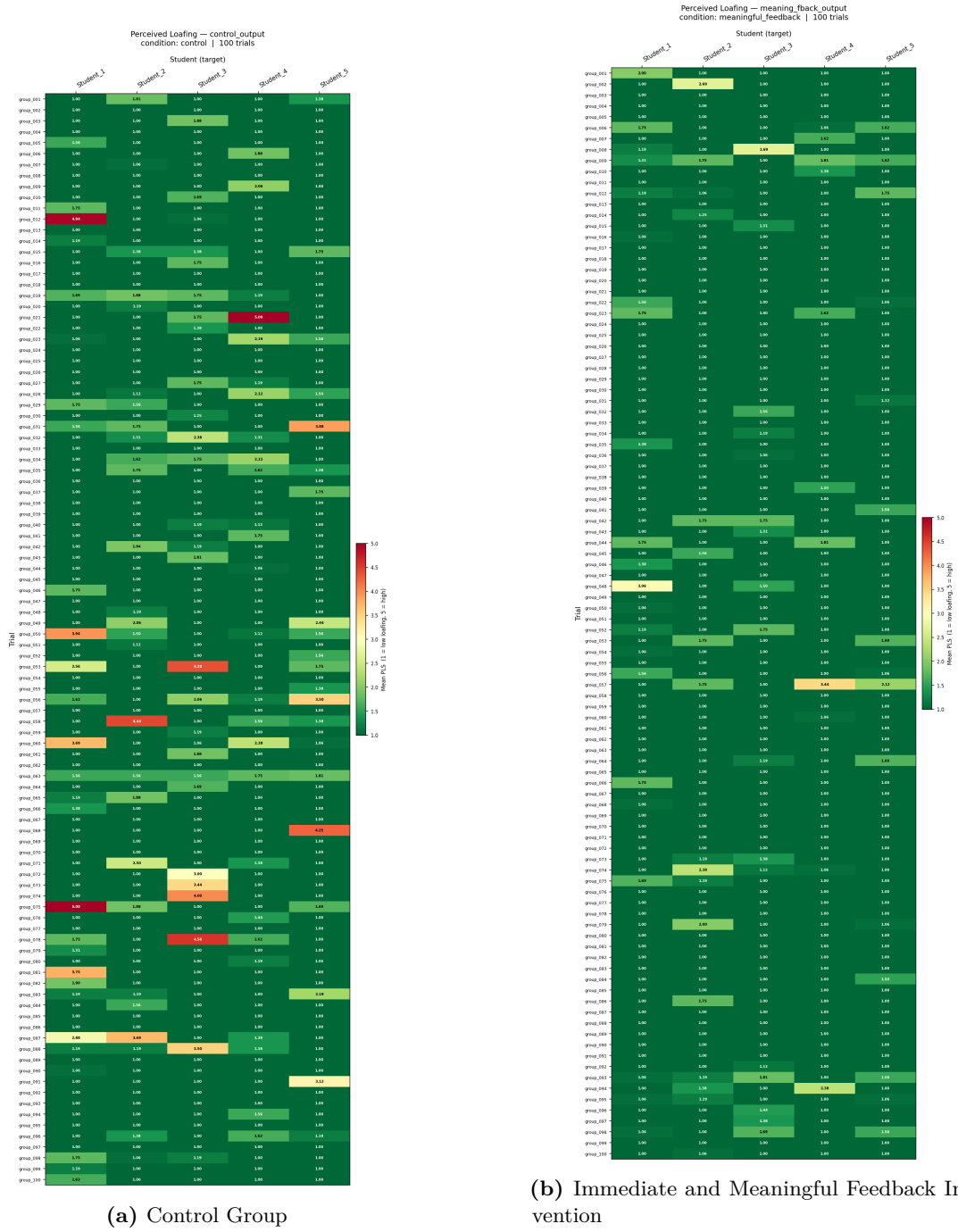


Figure 9.2. Control vs Immediate Meaningful Feedback Intervention

The immediate and meaningful feedback heatmap (Figure 9.2) is visually the most uniformly green of all intervention conditions. High-PLS cells are obsolete compared to the control, showing zero orange or red cells and very few yellow cells. This implies that immediate and meaningful feedback is an effective and promising intervention in mitigating

perceived social loafing.

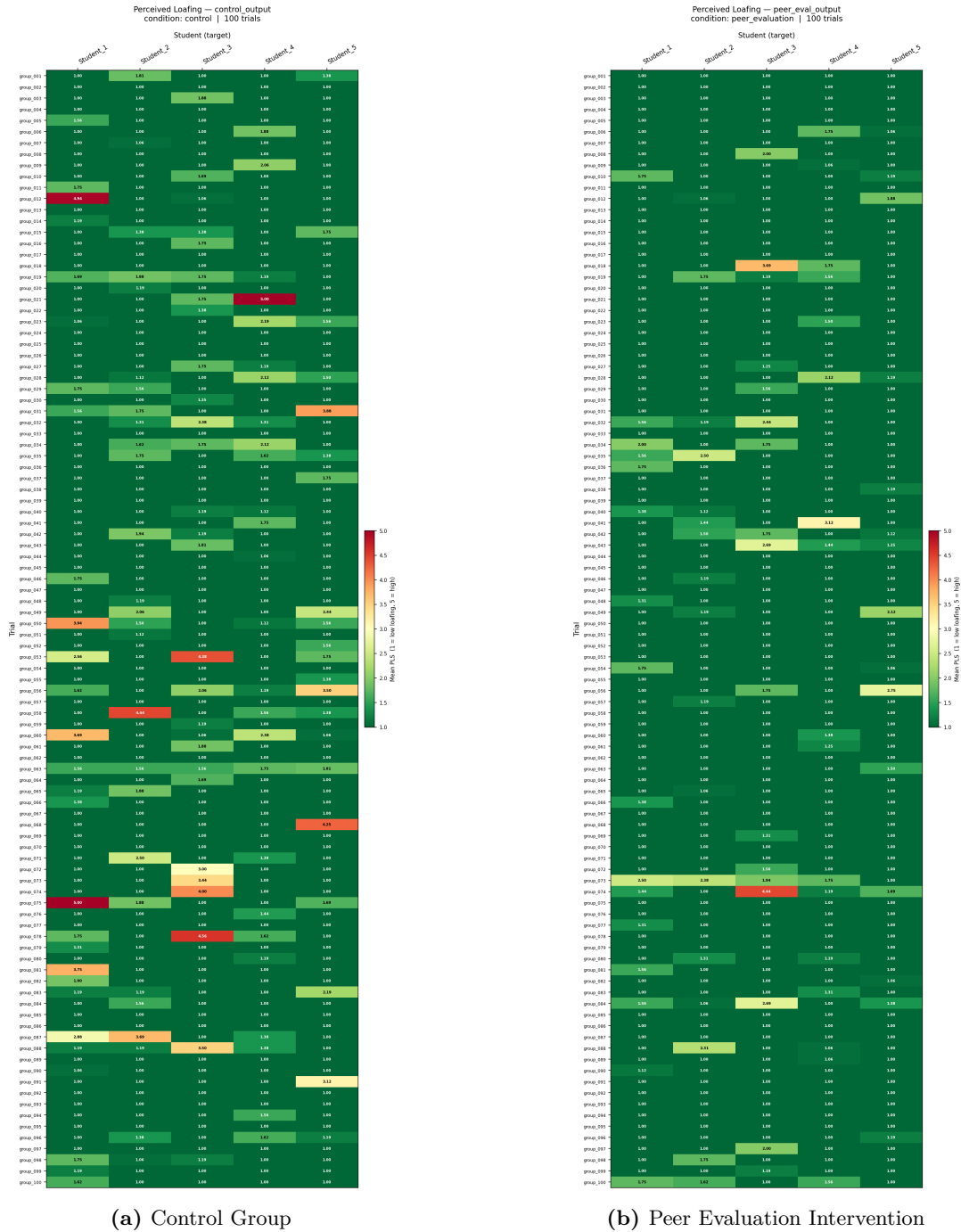


Figure 9.3. Control vs Peer Evaluation Intervention

The peer evaluation heatmap (Figure 9.3) shows a clear reduction in the frequency of elevated PLS scores relative to the control. The intervention condition has one orange cell and one red cell, with a few yellow cells, suggesting that the peer evaluation intervention is

moderately effective at mitigating perceived social loafing, yet still very much present.

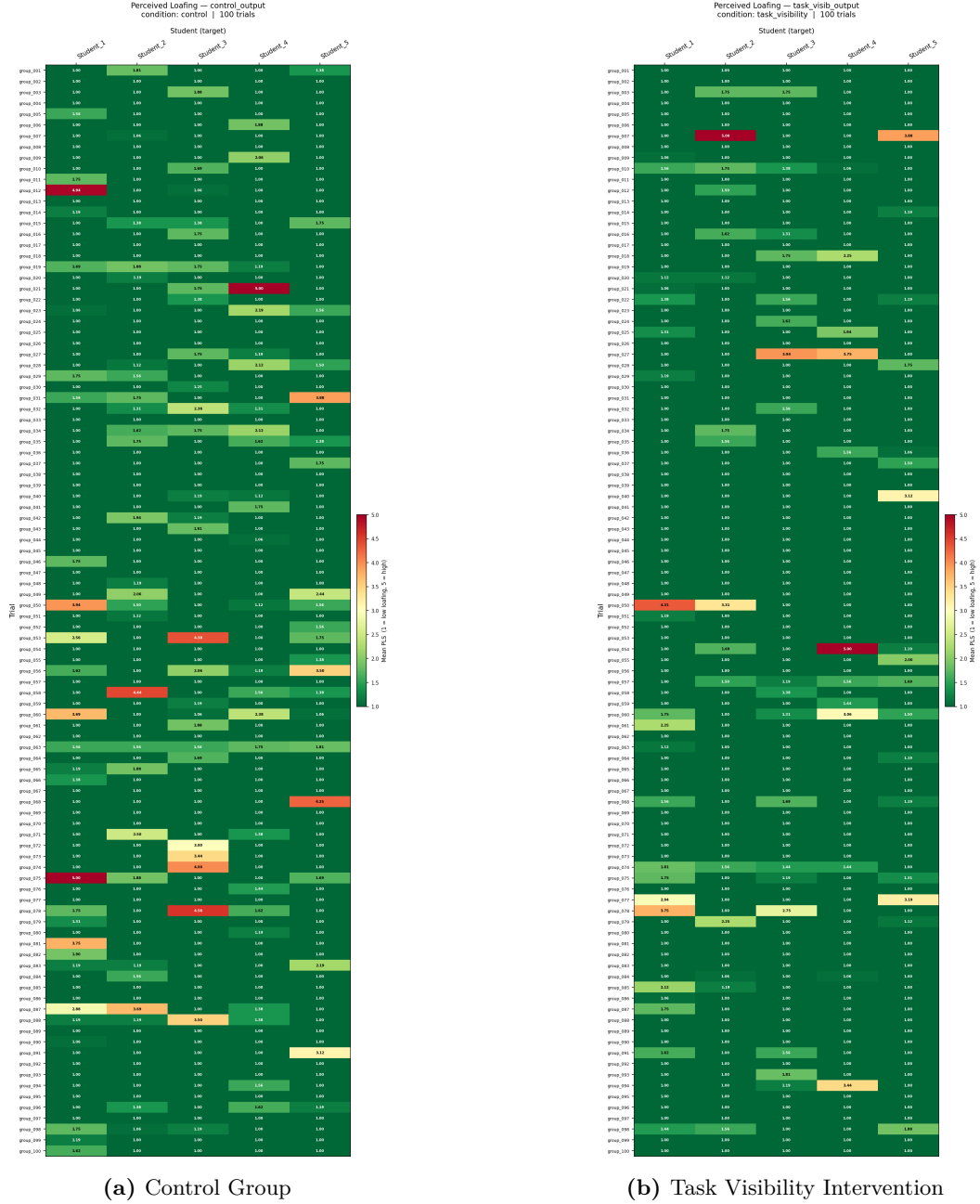
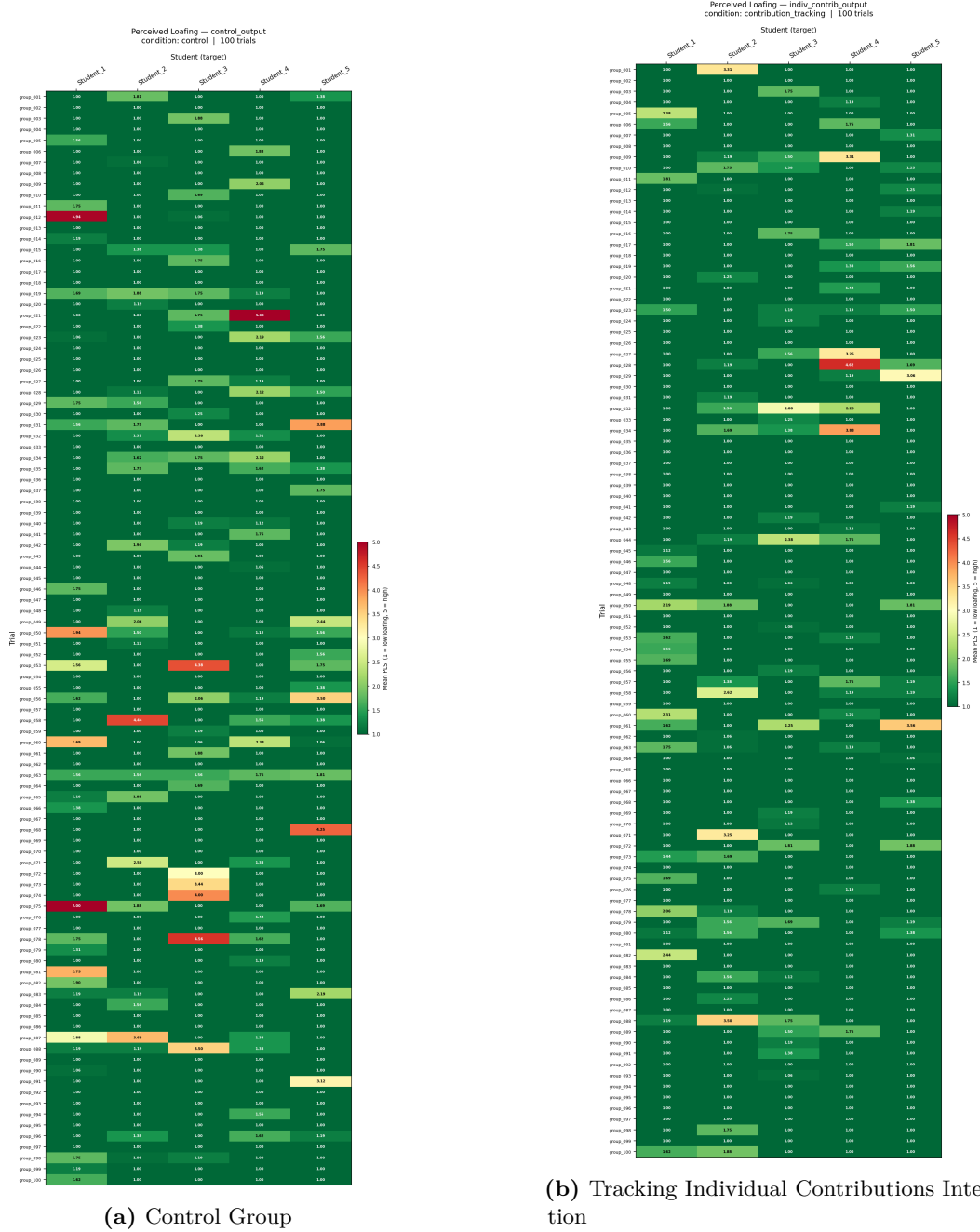


Figure 9.4. Control vs Task Visibility Intervention

The task visibility heatmap (Figure 9.4) shows a mixed picture. While the overall tone is greener than the control, isolated red cells remain in the intervention condition, indicating that task visibility reduces but does not eliminate high-PLS instances. There are still a considerable amount of red and orange cells in the intervention condition as compared to the control, suggesting that task visibility is a less effective intervention strategy.



(a) Control Group

(b) Tracking Individual Contributions Intervention

Figure 9.5. Control vs Tracking Individual Contributions Intervention

The individual contributions heatmap (Figure 9.5) shows a moderate reduction in high-PLS cells relative to the control, with fewer orange and red cells. Notably, there is only one red cell in the individual contributions heatmap, as compared to the four cells in the control, making the individual contributions intervention a moderate reducer of perceived social loafing.

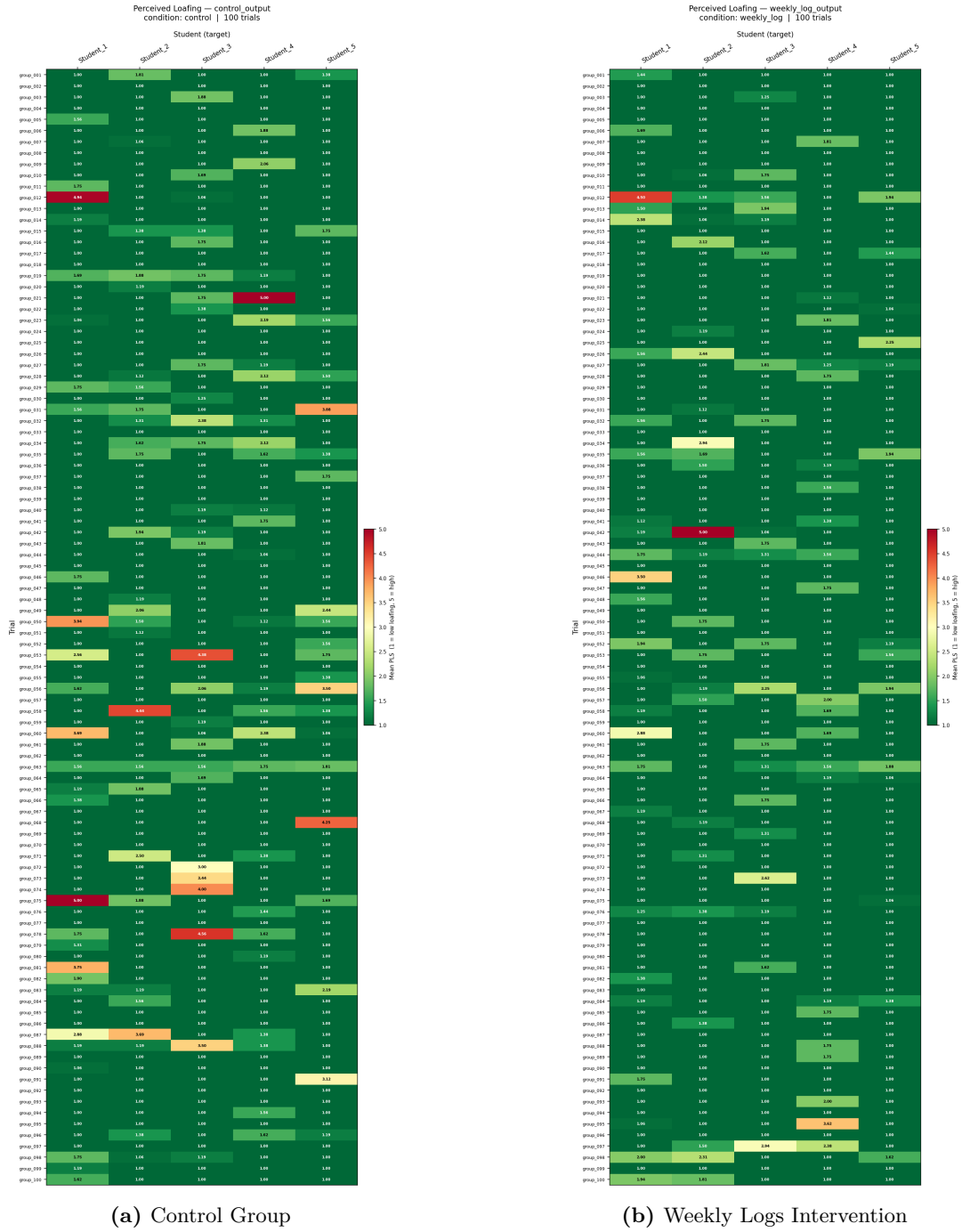


Figure 9.6. Control vs Weekly Logs Intervention

The weekly logs heatmap (Figure 9.6) shows a pattern similar to individual contributions, being mostly green, with one orange cell, one red cell, and a few yellow cells. Like the individual contributions intervention, the weekly logs are a moderate reducer of perceived social loafing.

9.2 Statistical Analysis

9.2.1 Dataset Overview

To formally compare perceived social loafing across conditions, an ordinary least squares (OLS) regression was estimated with PLS score as the outcome and condition as a categorical predictor, using the control condition as the reference category. Because observations are nested within groups (five students per group, 100 groups per condition) standard errors were clustered at the group level to account for the non-independence of observations within the same simulated team. All control-intervention contrasts between conditions were corrected for multiple comparisons using the Benjamini-Hochberg (BH) procedure to control the false discovery rate.

To reiterate, the direction of the PLS scale is: a higher PLS score indicates more perceived loafing, so negative regression coefficients and lower mean PLS values in the intervention conditions are the desired outcome.

Table 9.1. Dataset Overview by Condition

Condition	n_{obs}	n_{groups}	Mean PLS	SD PLS
Control	2000	100	1.236	0.644
Agile	2000	100	1.107	0.413
Immediate and meaningful feedback	2000	100	1.086	0.295
Peer evaluation	2000	100	1.104	0.364
Task visibility	2000	100	1.144	0.507
Tracking individual contributions	2000	100	1.149	0.446
Weekly log	2000	100	1.149	0.440

Table 9.1 shows that the control condition produced the highest mean PLS score (1.236), while meaningful feedback produced the lowest (1.086), followed closely by peer evaluation (1.104) and Agile (1.107). Notably, the control condition also had the highest standard deviation (0.644), indicating greater variability in perceived loafing across groups when no intervention was active. The meaningful feedback condition had the lowest standard deviation (0.295), suggesting that it not only reduced the average level of perceived loafing but also produced more consistent outcomes across teams.

9.2.2 OLS Regression with Cluster-Robust Standard Errors

Table 9.2. OLS Regression Results with Cluster-Robust Standard Errors (Reference: Control)

	Coef.	SE	z	p	95% CI low	95% CI high	Sig.
Intercept	1.2360	0.0302	40.98	<.001	1.1769	1.2951	***
Agile methodologies	−0.1290	0.0354	−3.64	<.001	−0.1984	−0.0596	***
Immediate and meaningful feedback	−0.1502	0.0339	−4.43	<.001	−0.2167	−0.0838	***
Peer evaluation	−0.1316	0.0327	−4.03	<.001	−0.1957	−0.0675	***
Task visibility	−0.0916	0.0381	−2.41	.016	−0.1663	−0.0170	*
Tracking individual contributions	−0.0870	0.0321	−2.71	.007	−0.1498	−0.0242	**
Weekly log	−0.0872	0.0336	−2.59	.010	−0.1532	−0.0213	*

$N = 14,000$; Adj. $R^2 = .010$; $F(6, 13993) = 4.139$, $p < .001$

Standard errors clustered at the group level.

* $p < .05$; ** $p < .01$; *** $p < .001$

9.2.3 Control-Intervention Contrasts with Benjamini-Hochberg Correction

Table 9.3. Control-Intervention Contrasts with Benjamini-Hochberg Correction ($n_{\text{sig}} = 6$ of 6)

Contrast	Est.	SE	CI low	CI high	t	p_{raw}	p_{BH}	Sig.
Control – Agile methodologies	0.1290	0.0354	0.0596	0.1984	3.64	<.001	.0020	*
Control – Immediate and meaningful feedback	0.1502	0.0339	0.0838	0.2167	4.43	<.001	.0002	*
Control – Peer evaluation	0.1316	0.0327	0.0675	0.1957	4.03	<.001	.0007	*
Control – Task visibility	0.0916	0.0381	0.0170	0.1663	2.41	.016	.0431	*
Control – Tracking individual contributions	0.0870	0.0321	0.0242	0.1498	2.71	.007	.0287	*
Control – Weekly log	0.0872	0.0336	0.0213	0.1532	2.59	.010	.0339	*

* $p_{\text{BH}} < .05$. Estimates reflect mean differences in PLS scores.

9.3 Hypothesis Testing

All six null hypotheses were rejected. Each intervention produced a statistically significant reduction in perceived social loafing relative to the control condition, as detailed below.

H1 - Agile. Agile methodology significantly reduced social loafing ($b = -0.129$,

$SE = 0.035$, $p < .001$, 95% CI $[-0.198, -0.060]$), supporting H_{A1} .

H2 - Immediate and Meaningful Feedback. Immediate and meaningful feedback produced the largest reduction in social loafing of all interventions tested ($b = -0.150$, $SE = 0.034$, $p < .001$, 95% CI $[-0.217, -0.084]$), supporting H_{A2} .

H3 - Peer Evaluation. Peer evaluation significantly reduced social loafing compared to the control condition ($b = -0.132$, $SE = 0.033$, $p < .001$, 95% CI $[-0.196, -0.068]$), supporting H_{A3} .

H4 - Task Visibility. Task visibility significantly reduced social loafing ($b = -0.092$, $SE = 0.038$, $p = .016$, 95% CI $[-0.166, -0.017]$), supporting H_{A4} .

H5 - Tracking Individual Contributions. Tracking individual contributions significantly reduced social loafing ($b = -0.087$, $SE = 0.032$, $p = .007$, 95% CI $[-0.150, -0.024]$), supporting H_{A5} .

H6 - Weekly Logs. Weekly logs significantly reduced social loafing ($b = -0.087$, $SE = 0.034$, $p = .010$, 95% CI $[-0.153, -0.021]$), supporting H_{A6} .

9.4 Intervention Rankings and Prioritization

Table 9.4 summarizes the interventions ranked by effect size.

Table 9.4. Interventions Ranked by Effect Size (Coefficient)

Rank	Intervention	Coefficient	p
1	Meaningful Feedback	-0.150	< .001
2	Peer Evaluation	-0.132	< .001
3	Agile	-0.129	< .001
4	Task Visibility	-0.092	.016
5	Weekly Logs	-0.087	.010
6	Individual Contributions	-0.087	.007

9.5 Interpretation

Benjamini-Hochberg corrected control-intervention contrasts revealed that all six null hypotheses were rejected. Every intervention produced a statistically-significant reduction in perceived loafing relative to control after Benjamini-Hochberg correction.

While all six interventions were significant, effect sizes varied meaningfully across condi-

tions. Meaningful feedback, peer evaluation, and Agile methodology produced the largest reductions in perceived loafing ($b = -0.150$, $b = -0.132$, and $b = -0.129$, respectively), while individual contributions, weekly logs, and task visibility produced smaller but still significant reductions (coefficients ranging from -0.087 to -0.092). The confidence intervals for the top three interventions show relatively little overlap with those of the bottom three, suggesting a meaningful gradient in effectiveness rather than a uniform effect across all conditions.

Meaningful feedback produced both the largest effect size and the narrowest confidence interval (95% CI [0.084, 0.217]) of any intervention tested, indicating that it not only reduced perceived loafing the most on average but also did so most consistently across simulated teams. Per Chapter 3, we interpret the ranking not the magnitude. Meaningful feedback ranks first; that is the basis for selecting it, not the -0.150 coefficient per se. This is corroborated by Table 9.1, which shows that the meaningful feedback condition had the lowest standard deviation of any condition ($SD = 0.295$), compared to $SD = 0.644$ in the control. Taken together, these findings identify meaningful feedback as the most promising intervention for implementation.

All six interventions produced statistically significant reductions in social loafing relative to the control condition, resulting in the rejection of all six null hypotheses. Among the interventions tested, immediate and meaningful feedback emerged as the most promising, producing the largest reduction in perceived loafing. Agile methodology and peer evaluation also demonstrated strong effects, and were statistically comparable to meaningful feedback. Task visibility, weekly logs, and individual contributions, while still significant, produced comparatively smaller and mutually indistinguishable effects.

These findings should be interpreted in light of the low overall R^2 . The interventions reliably shift the mean level of perceived loafing in the expected direction, but the amount of variance they explain is small. This is consistent with what the social loafing literature would predict: situational interventions matter, but so do individual differences in personality, motivation, and group composition, none of which are controlled by the interventions themselves. The practical takeaway is not that these interventions are weak, but that no single intervention can fully address a phenomenon as multifaceted as social loafing. Immediate and meaningful feedback is identified here as the strongest candidate for implementation, and its real-world effectiveness remains a question for future empirical work with human participants.

Chapter 10

From Prioritization to Implementation: Frank

The Concordia simulations identified meaningful feedback as the most promising intervention for mitigating social loafing in computer science group projects, performing significantly better than both weekly logs and tracking individual contributions in pairwise comparisons. The goal of the implementation phase is to turn this finding into a tool that fits naturally into the workflow of computer science teams to encourage immediate and meaningful feedback.

From these results, I created Frank: a Google Chrome extension that overlays the GitHub interface to promote immediate, specific, and actionable pull request feedback. Frank operates entirely within GitHub, which is a platform that computer science teams already use, and requires no changes to how a team manages its project. I designed Frank, not as a monitoring tool, but as a nudge, providing a lightweight intervention that raises the cost of low-effort feedback and makes the absence of timely review visible to the group. The purpose of Frank is to provide a step towards minimizing social loafing in computer science projects, while understanding that this tool alone will not eliminate social loafing.

10.1 Why a Tool, Not Just a Recommendation

The framework’s Step 5 output is a prioritized intervention: a hypothesis about what is most worth testing with human subjects. In principle, that output could be handed to instructors as a recommendation: “adopt immediate and meaningful feedback as a course norm” and the human-subject testing in Step 6 could proceed from there. In practice, this approach is unlikely to produce reliable evidence.

There are three reasons a tool is the better operationalization. The first is consistency. A norm communicated as a recommendation will be interpreted and implemented differently by different instructors and different teams. One instructor might enforce feedback through grading rubrics, another through verbal encouragement, another not at all. The resulting variance in implementation makes it nearly impossible to attribute downstream outcomes to the intervention itself rather than to differences in how it was deployed. A tool standardizes the intervention: every team using **Frank** gets the same feedback-quality checks and the same deadline visibility, which is a precondition for the randomized controlled trial that Step 6 calls for.

The second is friction. The social loafing literature consistently finds that interventions requiring significant changes to existing workflows have low adoption rates [20]. Asking instructors to introduce a new process, train students on it, and enforce it adds work to an already-full course. Asking students to internalize a behavioral norm without scaffolding is even less reliable. A tool that operates inside the platform teams already use lowers the cost of adoption to near zero on the team side and makes the intervention visible and persistent without requiring instructor enforcement.

The third is empirical leverage. A tool can collect structured data about whether the intervention is actually being followed, whether feedback is being given, whether it is timely, or whether it is substantive, in a way that a course-wide recommendation cannot. This data is essential for Step 6: without it, an RCT comparing treatment and control groups would have no way to confirm that the treatment was actually administered. *Frank* therefore plays a dual role: it is both the operationalization of the intervention and the instrumentation that makes its testing possible.

10.2 Why GitHub

The framework’s Step 2 selection criteria include implementability: an intervention should be feasible to deploy in the target setting. For computer science group projects, the target setting already has a shared platform that nearly all teams use to manage their work. GitHub hosts the code, the pull requests, the issue discussions, and the review comments that constitute most of the visible activity on a software project. Any intervention that lives on a different platform competes with GitHub for the team’s attention; an intervention that lives on GitHub itself does not.

Building *Frank* as a Chrome extension that overlays the existing GitHub interface preserves this property. Students do not need to install a separate application, learn a new workflow, or remember to visit a different site. The intervention appears where the work already happens. Pull request reviews are the natural moment at which feedback is given or withheld; placing a feedback-quality check at that moment is the most direct possible operationalization of the intervention the framework prioritized.

GitHub is also the right surface for the immediacy half of the intervention. Pull requests have observable timestamps, observable review activity, and observable state transitions, all accessible through the GitHub REST API. A live countdown badge on a pull request page is a feature that the platform’s existing structure makes possible without any custom server infrastructure on the researcher’s side. The platform does the heavy lifting; *Frank* only adds the nudge.

There is a tradeoff. Building on GitHub means that *Frank* cannot help teams that do not use GitHub, and cannot enforce its norms outside the pull request review surface. A

team that gives feedback verbally in a stand-up, or asynchronously in Slack, is invisible to the tool. The framework’s third selection criterion accepts this tradeoff: an intervention that fits naturally into the dominant platform is more valuable than one that aspires to comprehensive coverage but adds friction to teams that don’t already need it.

10.3 Technical Overview

Frank is implemented as a Manifest V3 Google Chrome extension. It consists of three main components: (1) a background service worker (`background.js`), (2) a content script (`content.js`), and (3) a popup interface (`popup.html` / `popup.js`).

10.3.1 Authentication and API Access

Frank authenticates with GitHub using a personal access token (classic) with `repo` and `gist` scope, provided by the user during setup. All GitHub communication is handled through the GitHub REST API (<https://api.github.com>), with the token sent as a `Bearer` header on every request. When saving a token, Frank immediately calls `GET /user` to validate the token and retrieve the user’s GitHub login, which is stored in `chrome.storage.sync` alongside the token and the configured stall threshold.

10.3.2 Data Storage

Group data, including members, open pull requests, feedback entries, and review deadlines, is stored in a private GitHub Gist as a single JSON file (`mf-data.json`). Each repository group maps to one Gist, and the mapping between repository keys and Gist IDs is stored locally in `chrome.storage.local`. To reduce API calls, Frank caches Gist data in `chrome.storage.local` with a two-minute expiry, falling back to a live fetch when the cache is stale or a forced refresh is requested. Gist writes use an optimistic retry strategy, so if a write fails, Frank retries up to three times with exponential backoff before causing an error.

10.3.3 Background Sync

Every 15 minutes, group data is refreshed and overdue pull requests are checked. If a pull request has passed its review deadline and doesn’t have a substantive review yet, Frank surfaces a Chrome desktop notification. Notifications are deduplicated using a flag stored in `chrome.storage.local`, so each overdue pull request triggers at most one notification per browser session.

10.3.4 Content Script

The content script is injected on all `github.com` pages and runs at `document_idle`. It monitors navigation events, including GitHub’s client-side routing via `history.pushState`

and `popstate`, and re-runs whenever the user navigates to a new pull request page. On a pull request page belonging to a Frank group, the content script makes sure feedback is immediate and substantive, described below.

10.4 Features

10.4.1 Specific, Actionable Feedback

A central finding in the social loafing literature is that meaningful feedback must be specific and actionable to be effective [20]. Frank enforces this by blocking pull request reviews that consist solely of low-effort responses. The following submissions are rejected by a regular expression matched against the trimmed review body:

- `lgtn`, `looks good`, `good`, `great`, `nice`
- `ok`, `okay`, `approved`
- `ship it`, `shipit`, `+1`
- Thumbs-up emoji (👍)
- Checkmark emoji (✓)

Frank intercepts review submissions at three levels to account for GitHub’s varied submission paths. First, it listens for `form` submit events in the capture phase. Second, it intercepts button click events on submit and review buttons. Third, because GitHub frequently posts comments and reviews via JavaScript without a traditional form submission, Frank patches `window.fetch` and `XMLHttpRequest.prototype.send` to intercept network-level POST requests to GitHub’s comment and review endpoints (`/comments`, `/reviews`, `/pull_request_reviews`). If the request body (parsed from JSON, URL-encoded form data, or `FormData`) follows the regular expression above of a low-effort response, the request is blocked and an inline warning is displayed above the text area prompting the reviewer to provide substantive feedback.

10.4.2 Immediate Feedback

Frank also addresses the need for *immediate*, meaningful feedback [20]. When a pull request is synced into a Frank group, a review deadline is computed by adding the group’s configured stall threshold (between 4 and 72 hours, set by the group creator in Settings) to the pull request’s opening timestamp. The content script injects a live countdown badge directly into the pull request page header, counting down in real time. The badge transitions through three visual states: green when time remains, amber when fewer than four hours remain, and a blinking red **OVERDUE** when the deadline has passed without a substantive review. Once

a substantive review is detected, via `GET /repos/{owner}/{repo}/pulls/{pr}/reviews`, filtered to exclude vague approvals, the badge displays a green checkmark and stops counting down. This creates a lightweight social accountability mechanism that is visible to all group members without requiring instructor involvement.

10.5 Setup

1. **Download the Chrome Extension.** Frank is available at https://chromewebstore.google.com/detail/kmdmiaiejpbpooacaijbbcbmpgmpocjen?utm_source=item-share-cb. Install it from the Chrome Web Store or load it as an unpacked extension via `chrome://extensions` in developer mode using <https://github.com/ashleygoh1001/frank>.

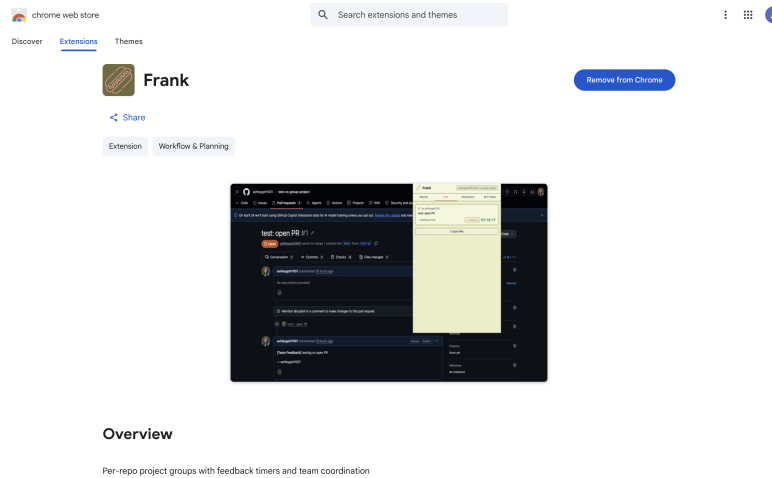


Figure 10.1. Downloading and installing the Frank Chrome extension.

2. **Create a GitHub personal access token.** Navigate to GitHub Settings → Developer Settings → Personal Access Tokens → Tokens (classic). Generate a new token with `repo` and `gist` scope.

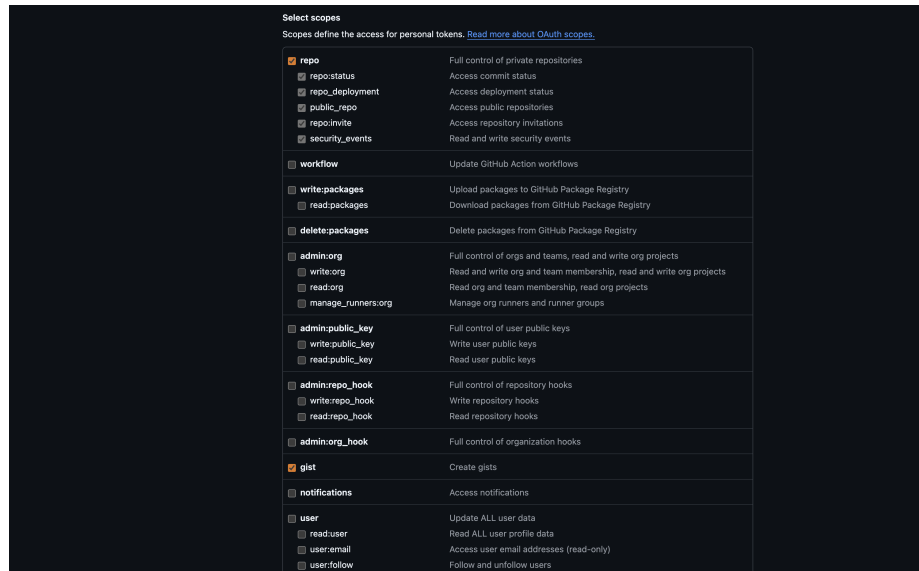


Figure 10.2. Generating a GitHub personal access token with `repo` and `gist` scope.

3. **Paste your token into Frank.** Open the Frank extension popup and enter your personal access token in the Settings tab. Frank validates the token against `GET /user` and stores your GitHub login for use in group operations.

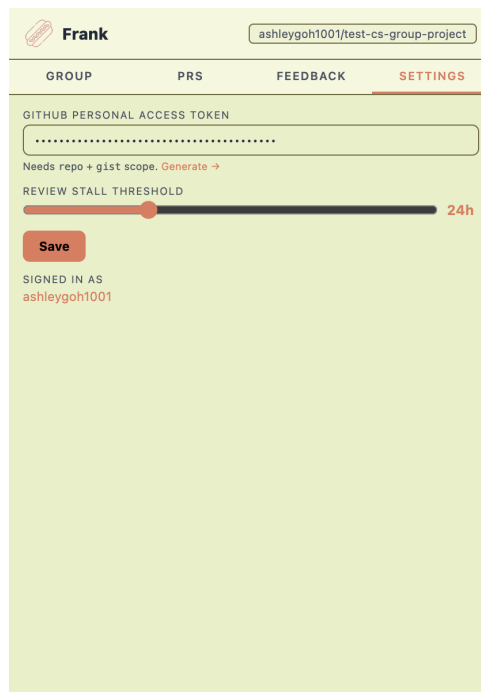


Figure 10.3. Entering a personal access token into Frank's Settings tab.

4. **Open the GitHub repository for your group project.** Navigate to the repository

you want to create a Frank group for. Frank detects the current repository from the page URL and displays it in the extension popup header.

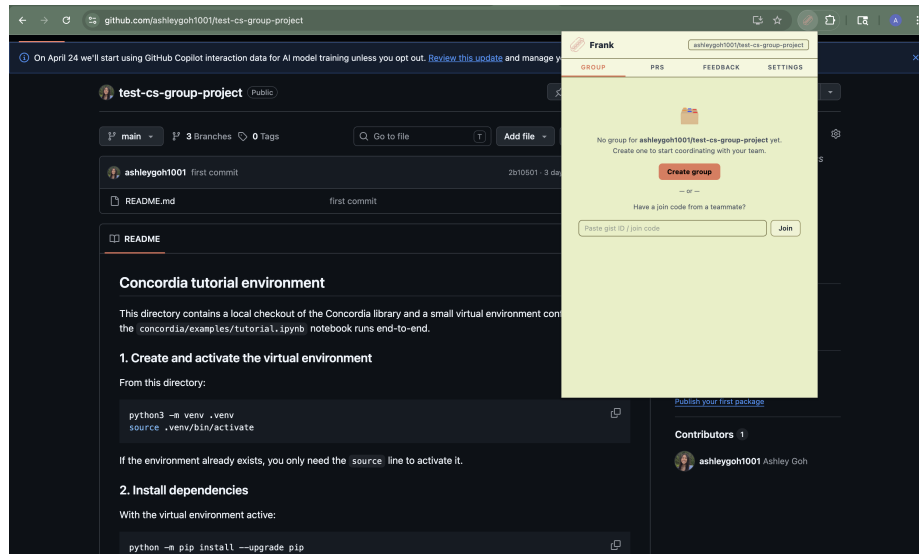


Figure 10.4. Navigating to the target GitHub repository.

5. **Create a group for the repository.** In the Group tab, click *Create group*. Frank creates a private GitHub Gist to store the group's data and generates a join code — the Gist ID — which is displayed in the popup and can be copied to the clipboard.

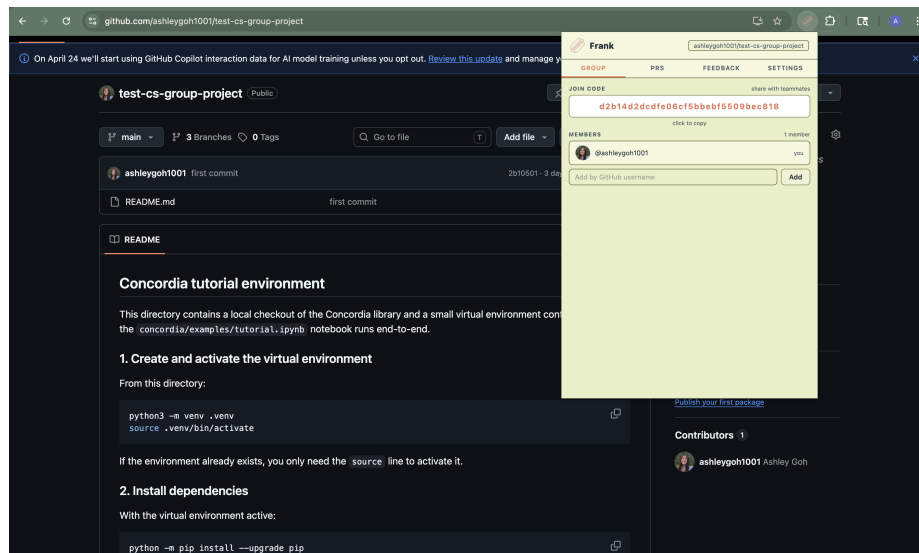


Figure 10.5. Creating a Frank group and copying the join code.

6. **Add your teammates.** Share the join code with teammates so they can paste it into the Group tab and join, or add them directly by GitHub username from within the

popup. Once added, Frank applies feedback quality checks and deadline visibility to all group members on the shared repository.

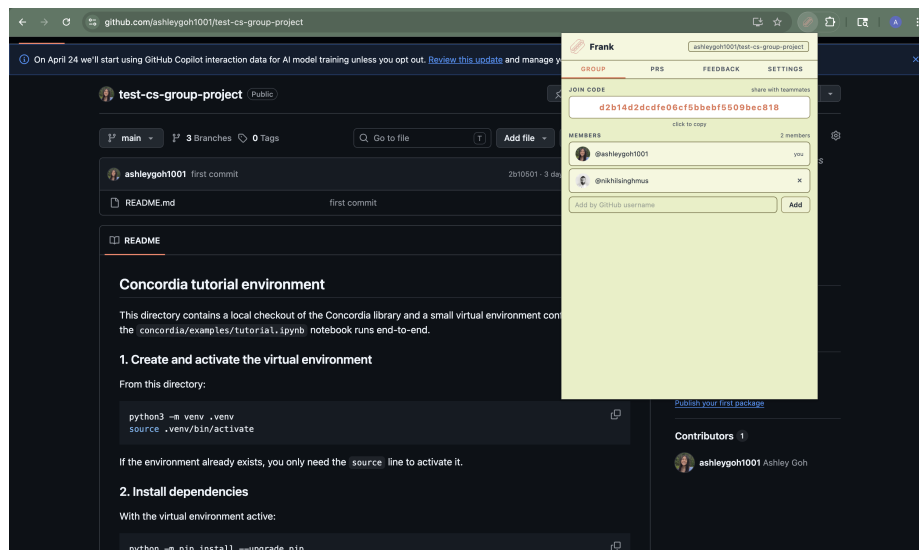


Figure 10.6. Adding teammates by join code or GitHub username.

10.6 Scope

Frank is intentionally scoped to what the framework's prioritization step identified as most effective: the immediate-and-meaningful-feedback intervention. It does not track individual contribution metrics, assign grades, or report data to instructors. Whether Frank produces measurable reductions in social loafing in real classrooms is the question that the framework's Step 6 (a randomized controlled trial) is designed to answer.

Part III

Reflection

Chapter 11

Limitations of the Case Study

11.1 Simulation Methodology

11.1.1 Prompt Engineering

To achieve realistic domain simulations, it is essential to align LLM agents with human knowledge and values through carefully designed prompts [5]. Effective alignment requires providing agents with task instructions, background knowledge, generation patterns, and domain-specific examples [5]. The prompts used in the Concordia simulations were not rigorously tested or optimized; a change in wording, or the insertion or deletion of a single phrase, can meaningfully alter agent behavior and simulation outcomes. The intervention and premise prompts did not undergo a sensitivity analysis. This introduces the possibility that the results reflect prompt design choices as much as the underlying interventions.

11.1.2 Single LLM

All simulations were run using Gemini 2.5 Flash as the underlying language model. No other LLMs were tested. Different models have different training data, alignment procedures, and behavioral tendencies, and it is possible that the relative ranking of interventions would differ under a different model.

11.1.3 Sample Size Relative to the Population

One hundred teams were randomly selected per condition from a theoretical population of over 17 quadrillion possible group combinations. While 100 trials is comparable to or larger than the participant pools used in most human-subject social loafing studies, it remains a small fraction of the possible space. It is important to understand that different random seeds could produce different team compositions and, potentially, different intervention rankings.

11.1.4 No Longitudinal Dynamics

Each simulation spans four weeks of a ten-week project, beginning at week 7. Social loafing often develops and compounds over time; some interventions may be effective early in a project but lose their effect as novelty wears off, while others may take time to produce behavioral change. By capturing only the latter portion of a project, the simulations may

not reflect how interventions perform across a full project arc. However, by choosing a specific span of time, more details and nuances of the interactions are captured.

11.1.5 No Persistent Social Context

Each of the 100 trials per condition is independent. Students have their own reputations, prior relationships, and shared histories that shape how they behave in groups. An agent who loafed in a previous project would face different social expectations in the next one. This accumulated social context is absent from the simulations, which may cause them to underestimate the social consequences of loafing over time.

11.2 Agent Modeling

11.2.1 Gender

Agents in the simulations were not assigned a gender. While prior research has found that gender affects social loafing (males are generally more likely to loaf than females), more recent work suggests that gender matters less than an individual’s view of their own self-uniqueness within a group [1]. Gender was omitted as a modeling dimension given this ambiguity, but its absence means the simulations do not capture any gender-related dynamics that may be present in real computer science group projects.

11.2.2 Culture

Agents were not modeled with cultural orientations. Research has shown that the cultural dimensions of individualism versus collectivism and power distance both influence an individual’s tendency to loaf: people from individualistic cultures are more inclined to loaf, while those from collectivistic cultures show more motivation in group settings [1]. The relationship between power distance and social loafing is particularly strong; researchers have called for further testing to determine whether individualistic cultural orientation causes loafing, or whether power distance alone is sufficient to predict it [1]. By omitting cultural dimensions from agent profiles, the simulations implicitly model a culturally homogeneous population, which limits the generalizability of the findings to diverse real-world classrooms.

11.2.3 Big Five Stability

The Big Five personality traits were used to differentiate agents, but the framework has known limitations. Trait rankings change with age and gender [25], meaning that the trait profiles assigned to simulated students may not accurately reflect the distribution of traits in a real undergraduate population. Additionally, the Big Five dimensions are correlated with cultural orientation [25], which further complicates their use in a simulation that does

not independently model culture. Isolating the effect of any single personality dimension on social loafing behavior is therefore difficult both in the simulations and in the literature on which the agent design is based.

11.3 Intervention Design

11.3.1 Interventions Tested in Isolation

Each of the six interventions were simulated independently, with one intervention active per condition. In practice, instructors commonly combine multiple strategies (for example, using Agile methodology alongside peer evaluation and individual contribution tracking simultaneously). The simulations provide no information about interaction effects between interventions, and it is possible that certain combinations would produce greater reductions in social loafing than any single intervention alone.

11.3.2 Intervention Fidelity

Each intervention was operationalized by injecting prompts at three levels of the simulation: the premise, the agent goal, and the situational context. This is an approximation of how interventions function in reality. Peer evaluation administered through a structured rubric in a real course involves social pressures, grading stakes, and interpersonal dynamics that cannot be fully captured by telling an agent it is being evaluated by its peers. The gap between prompt-level intervention and real-world implementation means that the simulation results may not perfectly predict how each intervention performs when applied to actual students.

Gaming the Intervention

A related but distinct limitation is that the simulations cannot capture strategic minimum-effort compliance with the interventions themselves. In the Concordia setup, each intervention is injected at three levels (the premise, the agent goal, and the situational context) which means that the intervention is a persistent feature of every agent’s reasoning on every step. An agent reasoning about whether or how to fill out a weekly log does so in a context where the weekly log requirement is part of its stated goal and immediate situation. The structural effect is that agents take the intervention seriously by construction.

Real students do not. Interventions that depend on student-generated artifacts are routinely gamed: weekly logs get filled with vague, recycled entries written five minutes before the deadline; peer evaluations get filled with uniform high scores to avoid conflict or to secure reciprocal high scores in return; contribution tracking boards get padded with trivial commits that inflate apparent activity without reflecting real work. In each case, the intervention’s form is satisfied while its function is hollowed out. The simulation cannot

reproduce this dynamic because the agents have no incentive to subvert a requirement that is built into their own reasoning context.

This matters for interpreting the case study’s results in two ways. First, the effect sizes reported in Chapter 9 are likely upper bounds for any intervention whose mechanism depends on substantive student engagement with a recurring artifact. Weekly logs, peer evaluations, and contribution tracking are the most vulnerable to this; Agile methodology and task visibility are somewhat less so because they depend more on group-level norms than on individual artifact production; immediate and meaningful feedback sits in between, since the feedback itself can be gamed (the “LGTM” problem that Frank is built to address) but the structural expectation of timely review is harder to circumvent without being conspicuous. Second, the ranking itself may be affected: an intervention that is easy to game in practice but hard to game in simulation will be over-ranked relative to one that resists gaming in both settings. This also contributes to the argument for treating the framework’s output as a filter rather than a final answer, and for closing the loop with the human-subject testing.

11.4 Measurement

11.4.1 Perceived Loafing as a Proxy

The Perceived Social Loafing Questionnaire measures how much group members perceive others to be loafing, rather than directly measuring actual effort reduction. While perceived loafing is a consequentially important outcome, it is what drives group dissatisfaction and performance loss regardless of whether actual loafing is occurring [28]. Perceived loafing is not a direct measure of behavior. Simulated agents completing a questionnaire about each other may reflect prompt-driven response patterns as much as genuine behavioral inference.

11.4.2 Questionnaire Limitations

Self-report and peer-report questionnaires are subject to well-documented limitations, including inaccurate understanding of questions, limited introspective ability, and response bias [25]. These limitations apply to the PSLQ as administered to LLM agents.

Chapter 12

Next Steps

12.1 Closing the Framework’s Loop: Testing Frank in a Real Classroom

While Frank was built upon social loafing literature and statistically significant results from simulations, Frank still needs to be tested in the real world. This would require a *randomized controlled trial* (RCT). For the RCT, project groups within the same course would be randomly assigned to either a treatment condition, in which they use Frank throughout the project, or a control condition, in which they use GitHub without Frank. From the RCT, we would have a better understanding of the actual effectiveness of Frank in a real-world situation.

12.2 Research-Backed Group Formation

A complementary direction for future work is to move backwards, starting at the very first point of group project, which is team formation. Rather than mitigating social loafing after groups have been formed, designing groups from the outset may reduce the likelihood of loafing emerging at all.

The simulation infrastructure developed for this thesis already models each agent along eight dimensions: the Big Five personality traits (openness, conscientiousness, extraversion, agreeableness, and neuroticism), preference for group work, winning orientation, and skill level. The social loafing literature provides hypotheses about which combinations of these traits are more or less conducive to loafing. For example, groups with higher average conscientiousness and winning orientation tend to loaf less, while groups with low cohesiveness and high perceived dispensability of effort tend to loaf more [30, 38].

A natural extension of this work would be to develop a group formation tool that assigns students to project groups based on these dimensions, with the goal of composing teams whose trait profiles are predicted to minimize social loafing. For example, students could complete a short survey at the start of a course to measure the Big Five, preference for group work, winning orientation, and self-assessed skill level. Then, an algorithm could use these inputs to form balanced or strategically complementary groups.

12.3 Extending Frank

Frank currently implements a single intervention (meaningful feedback) because that is what the simulations identified as most promising. However, the control-intervention contrasts showed that Agile methodology and peer evaluation were statistically comparable to meaningful feedback and significantly outperformed the control. A natural extension of Frank would be to implement these additional interventions within the same GitHub interface.

Peer evaluation, for example, could be implemented as a structured in-extension survey that prompts group members to rate each other on contribution, communication, and reliability at configurable intervals, mirroring the multi-point, criteria-specific peer evaluation protocol used in the Concordia simulations. Task visibility could be implemented by surfacing a shared task board within the Frank overlay, where each task has a named owner and a visible deadline. Each of these features would extend Frank's reach from a single-intervention tool into a more comprehensive platform for mitigating social loafing across multiple dimensions simultaneously. As Frank is extended, the interaction effects between interventions (which the simulations were unable to test) become an important empirical question.

Chapter 13

Conclusion

This thesis proposed that generative agent-based models can serve as a prioritization layer in the social scientist’s workflow as a step inserted between literature review and human-subject experimentation that uses simulation to narrow a large set of candidate interventions down to the one or few most worth testing with real participants. The six-step framework is the durable contribution of this thesis; the case study on social loafing in computer science group projects is the evidence that it works end-to-end.

The framework’s claim rests on a specific epistemic stance. Drawing on Epstein’s generative social science, Wu et al.’s alignment problem, and Lu et al.’s distinction between believability and accuracy, the framework treats simulation output as a filter rather than an oracle. It does not predict real-world effect sizes; it produces a rank ordering of interventions, and the highest-ranked intervention is not claimed to be the most effective in practice; it is simply the one most worth testing first. The framework is designed around the weaker, more defensible claim that LLM-based simulations can support ordinal comparison even when they cannot support effect-size estimation.

The case study was a proof of concept that the framework operates end-to-end. A systematic review surfaced six empirically supported candidate interventions for social loafing in computer science group projects. Using Concordia and Gemini 2.5 Flash, 700 simulations were run across seven conditions, with agents heterogeneous along eight dimensions drawn from the social loafing literature. All six interventions produced statistically significant reductions in perceived social loafing, with meaningful feedback ranking first. That ranking was then translated into Frank, a Chrome extension that operationalizes the prioritized intervention inside an existing student workflow — closing the loop from literature to simulation to deployable tool. Whether Frank produces measurable reductions in social loafing with real students is the open question for the framework’s sixth step.

The broader claim is methodological. The framework is portable to any domain where multiple plausible interventions exist, the relevant individual differences are identified by the literature, and the cost of comparing those interventions empirically is high. In such domains, a heterogeneous, scenario-level simulation can direct human-subject resources toward the interventions most likely to repay them. This shifts the prioritization decision from intuition to evidence, even when that evidence is itself imperfect, and it makes broad comparative research feasible for problems where it previously was not. The framework’s value lies not in replacing human-subject studies, but in making sure the human-subject studies that do get

run are aimed at the candidates most likely to produce meaningful results. More broadly still, the framework’s machinery is not limited to ranking interventions the literature has already named: the same heterogeneous-agent, comparative-evaluation pipeline can be used to surface hypotheses the literature has not yet articulated, with the simulation acting as a filter on candidate questions rather than candidate answers.

Generative agent-based models will not settle social science questions on their own, and this thesis does not claim that they should. What they can do, used carefully, is change the shape of the workflow that surrounds those questions by narrowing the field of candidates, surfacing unanticipated dynamics, and lowering the cost of asking comparative questions in the first place. The framework presented here is one attempt to make that role concrete. The case study is a demonstration that the workflow runs. The next step, for this domain and others, is to run it again.

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Full Intervention Prompts

.1 Agile Methodologies

Level 1: Premise

This team has fully adopted an Agile philosophy from week 1. Work is organized in weekly sprints with a sprint backlog, a daily stand-up, a sprint review, and a retrospective. The Agile norm emphasizes collective ownership, mutual support, and transparency: team members are expected to surface blockers immediately in stand-ups rather than hiding problems, to help teammates who are stuck rather than working in isolation, and to hold each other accountable through retrospectives.

Today's session is a sprint planning + stand-up for sprint 3.

— CURRENT SPRINT BACKLOG (Week 7, Sprint 3) —

Sprint goal: finalize architecture and assign coding modules.

Backlog items (owner TBD at today's sprint planning):

- [ARCH-01] Agree on system architecture | Story points: 3
- [ARCH-02] Document architecture decision in README | Story points: 1
- [CODE-01] Scaffold module structure in repository | Story points: 2
- [CODE-02] Assign a coding module to each team member | Story points: 1
- [TEST-01] Define acceptance criteria for sprint 3 | Story points: 2

Sprint 2 retrospective action items carried forward:

- Improve response time on PR reviews (from retrospective)
- Assign clear owners before coding starts (from retrospective)

— END SPRINT BACKLOG —

Level 2: Agent Goal

Your team follows an Agile methodology with weekly sprints, stand-ups, and retrospectives. The Agile norm means you support your teammates openly rather than hiding problems, you raise blockers immediately, and you hold and are held to collective commitments. Today is sprint planning for sprint 3.

Level 3: Agent Situational Context

The team is running Agile sprints with stand-ups, sprint reviews, and retrospectives throughout the project. The Agile philosophy means supporting teammates, surfacing problems immediately, and taking collective ownership of the sprint goal rather than working in isolation or concealing difficulties.

— CURRENT SPRINT BACKLOG (Week 7, Sprint 3) —

Sprint goal: finalize architecture and assign coding modules.

Backlog items (owner TBD at today's sprint planning):

- [ARCH-01] Agree on system architecture | Story points: 3
- [ARCH-02] Document architecture decision in README | Story points: 1
- [CODE-01] Scaffold module structure in repository | Story points: 2
- [CODE-02] Assign a coding module to each team member | Story points: 1
- [TEST-01] Define acceptance criteria for sprint 3 | Story points: 2

Sprint 2 retrospective action items carried forward:

- Improve response time on PR reviews (from retrospective)
- Assign clear owners before coding starts (from retrospective)

— END SPRINT BACKLOG —

Game Master

INTERVENTION ACTIVE: Agile Methodology. The team runs weekly sprints with stand-ups, reviews, and retrospectives. The core Agile norm in force is mutual support and transparency: team members are expected to surface blockers immediately rather than hiding them, to help stuck teammates, and to take collective ownership of the sprint goal.

Today is sprint 3 planning. When narrating, reflect Agile rhythms: agents give stand-up updates, negotiate sprint backlog ownership, and surface retrospective action items. Social loafing should be harder to sustain because the Agile structure makes it visible, though personality differences still affect how well individuals embrace the philosophy.

.2 Immediate and Meaningful Feedback

Level 1: Premise

The course staff and team have agreed that every contribution receives meaningful and timely feedback. Concretely:

- Code submitted to the shared repository is reviewed within 24 hours with

specific, actionable comments (not just 'LGTM').

- Design proposals are discussed in the same session they are raised, with a clear decision or next step recorded.
- When a teammate is struggling or off-track, the team addresses it directly and constructively in the current session rather than letting it accumulate.
- Course staff provide written feedback on each weekly artifact within 48 hours. This norm is in effect for the full duration of the project.

Level 2: Agent Goal

The team norm requires that every contribution you make receives meaningful and timely feedback, and that you provide the same to your teammates. Do not leave code reviews, design decisions, or teammate concerns unaddressed until the next session—respond specifically and promptly within the current session.

Level 3: Agent Situational Context

A meaningful-and-immediate-feedback norm is active throughout this project. Every contribution — code, design, documentation — receives specific and timely feedback. Concerns about teammate performance are raised and addressed directly in the same session they arise.

Game Master

INTERVENTION ACTIVE: Meaningful and Immediate Feedback. The team norm requires specific, timely feedback on all contributions within the same session or within 24 hours for async work. When narrating, reflect that agents do not silently ignore each other's work or problems. Design proposals get real-time responses. Code is reviewed specifically and promptly. If a teammate is struggling, others address it directly and constructively this session. Feedback is substantive — not vague praise or silent acceptance.

.3 Peer Evaluation

Level 1: Premise

The course uses a structured peer-evaluation system implemented every week. Week 7, 8, 9, and 10 evaluations are still to be completed. Each round uses specific evaluative criteria: contribution to tasks, communication quality, and reliability (meeting commitments). Scores are averaged across teammates and returned to each student with written comments. They contribute to individual

project grades.

— PEER EVALUATION ROUND 6 (Week 6, on file) —

Criteria rated 1-5 by teammates (average scores):

Student_1: contribution=?, communication=?, reliability=? [entries pending]

Student_2: contribution=?, communication=?, reliability=? [entries pending]

Student_3: contribution=?, communication=?, reliability=? [entries pending]

Student_4: contribution=?, communication=?, reliability=? [entries pending]

Student_5: contribution=?, communication=?, reliability=? [entries pending]

Week 7 peer evaluations are due at the end of week 7.

— END EVALUATION ROUND 6 —

Level 2: Agent Goal

Your performance is evaluated by your teammates at multiple points during the project using specific criteria: task contribution, communication quality, and reliability. Six evaluation rounds have already occurred; the next is at the end of week 7 (this week). How you behave in every session affects how your peers will rate you.

Level 3: Agent Situational Context

Online peer evaluations using specific criteria (contribution, communication, reliability) have been running since week 1 and continue through week 10. Your teammates are observing and will evaluate your behavior in this session.

Game Master

INTERVENTION ACTIVE: Online Peer Evaluation (early, multi-point, criteria-specific). Agents know they are being evaluated by peers on contribution, communication, and reliability. Two rounds have passed; one remains. When narrating events, reflect that this accountability shapes behavior: agents are aware that unhelpful, unreliable, or uncommunicative conduct will be reflected in peer scores. Surface realistic social dynamics — including strategic impression management — that arise from peer accountability.

.4 Task Visibility

Level 1: Premise

Before this meeting the team agreed on three structural norms: (1) Performance targets — every task has a named owner, a clear quality definition, and a firm deadline visible to all. (2) Communication procedures — team members post a brief update whenever they start, finish, or are blocked on a task. (3) Problem-solving methods — when a blocker arises, the owner flags it immediately and the team uses a defined escalation path (pair-help → group discussion → instructor) rather than staying stuck silently. These norms are in effect for the remainder of the project.

Level 2: Agent Goal

Your team operates under explicit task-visibility norms: every task has a named owner, a performance target, and a deadline that everyone can see. You are expected to post updates when you start, finish, or are blocked, and to follow the agreed escalation path when problems arise. Work within these norms throughout the project.

Level 3: Agent Situational Context

The team has established task-visibility norms that are active throughout this project: explicit performance targets, communication update procedures, and an agreed problem-solving escalation path. Every task must have a named owner and a visible deadline.

Game Master

INTERVENTION ACTIVE: Task Visibility — performance targets, communication procedures, and problem-solving methods. When narrating events, reflect that the team operates under explicit structural norms. Agents know who owns which task and what the quality target is. Updates are expected when work changes state. Blocked agents are expected to escalate rather than go silent. Personality differences still apply.

.5 Tracking Individual Contributions

Level 1: Premise

Each team member’s contributions are logged on a shared board visible to all teammates and to course staff. The board records design decisions, code commits, test coverage, and documentation written by each person. An up-to-date entry is required from every member after each session. Visible contributions recognize effort and achievement; blank or sparse entries are equally visible and will be noted.

Level 2: Agent Goal

Your individual contributions to this project are tracked on a shared board that your teammates and course staff can see at any time. Whatever you personally design, code, test, or document will appear next to your name. Making meaningful, visible contributions matters for your sense of personal achievement and for your grade.

Level 3: Agent Situational Context

A shared individual contribution board is active throughout this project. Every member’s work is logged and publicly visible to the team and to course staff after each session. Your contribution entry for this session must reflect what you personally accomplished.

Game Master

INTERVENTION ACTIVE: Individual Contribution Tracking. Every agent’s work is logged on a shared board visible to all teammates and to course staff. When narrating events, reflect that agents are aware their individual contributions—or lack of them—are transparent. Agents who contribute feel a sense of personal achievement; agents who avoid tasks know their blank entry will be noticed. Discussions about who does what should be shaped by this visibility. The board is updated at the end of this session.

.6 Weekly Logs

Level 1: Premise

INTERVENTION — WEEKLY PROGRESS LOG REQUIREMENT:
INTERVENTION — WEEKLY PROGRESS LOG:

The course requires the team to submit a structured weekly progress log every week. Each log records: (1) project milestones and whether they were met; (2) each team member's individual progress and contributions that week; and (3) expected completion dates for all outstanding tasks. Logs are reviewed by course staff and are part of your grade. Incomplete or missing individual entries are flagged to the instructor.

Logs from weeks 1 through 6 are already on file. The week 7 log is due at the end of today's meeting.

— WEEK 6 PROGRESS LOG (on file) —

Milestones this week:

- System requirements draft: MISSED (still in discussion)
- GitHub repository setup: COMPLETE
- Architecture decision: MISSED (pending team agreement)

Individual progress:

- [entries varied; some members submitted brief notes, others left blank]

Outstanding tasks and expected completion dates:

- Finalize architecture: target week 7
- Assign coding modules to team members: target week 7
- Set up CI/CD pipeline: target week 8
- Core feature implementation: target weeks 8-9
- Integration and bug fixing: target week 9
- Final demo preparation: target week 10

— END WEEK 6 LOG —

Level 2: Agent Goal

Your team submits a structured weekly progress log every week. The week 7 log—covering milestones, your personal contributions, and updated completion dates—is due at the end of today's meeting. Your name will appear alongside your entry; a blank entry is visible to your teammates and to course staff.

Level 3: Agent Situational Context

A structured weekly progress log is required throughout this project. Every log covers: (1) milestones met or missed, (2) individual contributions per team

member, and (3) updated expected completion dates. Logs are reviewed by course staff. The week 7 log is due at the end of this meeting.

— WEEK 6 PROGRESS LOG (on file) —

Milestones this week:

- System requirements draft: MISSED (still in discussion)
- GitHub repository setup: COMPLETE
- Architecture decision: MISSED (pending team agreement)

Individual progress:

- [entries varied; some members submitted brief notes, others left blank]

Outstanding tasks and expected completion dates:

- Finalize architecture: target week 7
- Assign coding modules to team members: target week 7
- Set up CI/CD pipeline: target week 8
- Core feature implementation: target weeks 8-9
- Integration and bug fixing: target week 9
- Final demo preparation: target week 10

— END WEEK 6 LOG —

Game Master

INTERVENTION ACTIVE: Weekly Progress Log requirement. The team must submit a weekly log covering milestones, individual member contributions, and completion dates. Logs for weeks 1-6 are on file. The week 7 log is due by the end of this session. Week 6 log notes: architecture still pending; task assignments not formally agreed; contributions uneven. When narrating, reflect that the log creates concrete accountability: agents know they must report what they personally accomplished. A blank entry next to a name is visible to everyone. Task and deadline decisions made in this meeting become the milestone and completion-date entries that the team is held to next week.